

Matrices

1.1. REVIEW OF FUNDAMENTAL CONCEPTS AND ALGEBRA OF MATRICES

(i) **Definition of Matrices** : A set of mn numbers (real or complex) arranged in a rectangular array having m rows (horizontal lines) and n columns (vertical lines), the numbers being enclosed by brackets $[]$ or $()$ is called an $m \times n$ matrix (read as "m and n" matrix). An $m \times n$ is usually written as

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ a_{41} & a_{42} & a_{43} & \dots & a_{4n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

Here each element has two suffixes. The first suffix indicates

the row and second suffix indicates the column in which element lies.

(ii) **Square Matrix** : A matrix in which number of rows is equal to the number of columns is called a square matrix.

(iii) **Multiplication of a Matrix by a Scalar** : When each element of a matrix A (say) is multiplied by a scalar k (say) then kA is defined as multiplication of A by a scalar k .

(iv) **Matrix Multiplication** : Two matrices A and B are said to be conformable for the product AB if number of columns of A is equal to the number of rows of B .

Thus if $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{n \times p}$ where $1 \leq i \leq m$, $1 \leq j \leq n$, $1 \leq k \leq p$. Then AB is defined as the matrix $C = [c_{ik}]_{m \times p}$

where $c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + a_{i3}b_{3k} + \dots + a_{in}b_{nk}$

$$= \sum_{j=1}^n a_{ij}b_{jk}$$

(v) **Properties of Matrix Multiplication** :

(a) Matrix multiplication is not commutative in general i.e., $AB \neq BA$

(b) Matrix multiplication is associative i.e., $A(BC) = (AB)C$.

(c) Matrix multiplication is distributive w.r.t. matrix addition i.e., $A(B+C) = AB+AC$

(d) If A, I are square matrices of the same order then $AI = IA = A$

(e) If A is a square matrix of order n then $A \times A = A^2$, $A \times A \times A = A^3$, $A \times A \times A \times \dots \times A$ m times $= A^m$

(vi) **Transpose of a Matrix** : Given a matrix A then matrix obtained from A by changing its rows into columns and columns into rows is called the transpose of A and is denoted by A' or A^t .

(vii) **Properties of Transpose of Matrix** :

(a) $(A')' = A$,

(b) $(A+B)' = A' + B'$

(c) $(AB)' = B'A'$ known as **Reversal Law of transposes**.

(viii) (a) **Symmetric Matrix** : A square matrix is said to be symmetric if $A' = A$ i.e., if $A = [a_{ij}]$ then $a_{ij} = a_{ji} \forall i, j$

(b) **Skew symmetric Matrix** : A square matrix is said to be skew symmetric if $A' = -A$ i.e., if $A = [a_{ij}]$ then $a_{ij} = -a_{ji} \forall i, j$ and when $i = j$. Then $a_{ii} = 0$ for all values of i .

Thus in a skew symmetric matrix all diagonal elements are zero.

(ix) **Involutory Matrix** : A square matrix A is said to be involutory if $A^2 = I$

(x) **Orthogonal Matrix** : A square matrix is called an orthogonal matrix if $AA' = A'A = I$

(xi) **Complex Matrix** : A matrix is said to be a complex matrix if at least one element of the matrix is a complex number $a + ib$ where a, b are real.

(xii) (a) **Conjugate of a Matrix** : If in a complex matrix A the elements are replaced by their conjugates then matrix so obtained is called conjugate of a matrix and is denoted by $\bar{A}$.

(b) **Conjugate Transpose of a Matrix** : A matrix A obtained by replacing the elements by their conjugates and then taking transpose of the new matrix is called conjugate transpose of the matrix and is denoted by $(\bar{A})' = A^\theta$

(xiii) (a) **Hermitian Matrix** : A square matrix A is said to be hermitian if $A^\theta = A$ i.e., if $A = [a_{ij}]$ then $a_{ij} = a_{ji} \forall i, j$

$\therefore$ When $i = j$, $\bar{a}_{ii} = a_{ii} \Rightarrow a_{ii}$ is purely real

$\therefore$ In hermitian matrix all diagonal elements are purely real.

(b) **Skew Hermitian Matrix** : A square matrix A is said to be skew Hermitian if $A^\theta = -A$ i.e., if $A = [a_{ij}]$ then $\bar{a}_{ij} = -a_{ji} \forall i, j$. When $i = j$ then $\bar{a}_{ii} = -a_{ii} \Rightarrow a_{ii}$ is purely imaginary or zero.

$\therefore$ In a skew Hermitian matrix diagonal elements are purely imaginary or zero

(xiv) **Unitary Matrix** : A square matrix A is called a unitary matrix if $A^\theta A = A A^\theta = I$.

(xv) **Adjoint of a Square Matrix** : The adjoint of a square matrix is the transpose of the matrix obtained by replacing each element of A by its co-factors in $|A|$

$$A (\text{Adj } A) = (\text{Adj } A) A = |A| I_n; n \text{ being the order of matrix } A$$

(xvi) **Singular and Non-Singular Matrices** : A square matrix is said to be singular if $|A| = 0$ and non-singular if $|A| \neq 0$.

(xvii) (a) **Inverse of a Square Matrix** : Let A be a square matrix of order n . If there exists another matrix B of the same order such that $AB = BA = I$ then matrix A is said to be invertible and B is called inverse of A . Inverse of A denoted by A^{-1} . Thus $B = A^{-1}$ and $A A^{-1} = A^{-1} A = I$. From (xv) we see that $A (\text{Adj } A) = |A| I$

$$\therefore A^{-1} = \frac{\text{Adj } A}{|A|}$$

(b) The inverse of a square matrix, if it exists, is unique.

(c) The necessary and sufficient condition for a square matrix A to possess inverse is that $|A| \neq 0$ i.e., A is non singular.

(d) If A is invertible, then so is A^{-1} and $(A^{-1})^{-1} = A$

(xviii) If A and B are two non-singular matrices of the same order, the $(AB)^{-1} = B^{-1} A^{-1}$ which is known as **Reversal Law of Inverses**.

(xix) If A, B are two square matrices of the same order, then $\text{Adj } (AB) = (\text{Adj } B) (\text{Adj } A)$ known as **Reversal Law of Adjoints**.

1.2. ELEMENTARY TRANSFORMATIONS (OR OPERATIONS)

(P.T.U.)

Anyone of the following operations on a matrix is called an elementary transformation (or E-operation).

(i) **Interchange of two rows or columns.**

The interchange of i th and j th rows is denoted by R_{ij} .

The interchange of i th and j th columns is denoted by C_{ij} .

(ii) **Multiplication of (each element of) a row or column by a non-zero number k .**

The multiplication of i th row by k is denoted by kR_i .

The multiplication of i th column by k is denoted by kC_i .

(iii) **Addition of k times the elements of a row (column) to the corresponding elements of a (or column), $k \neq 0$.**

The addition of k times the j th row to the i th row is denoted by $R_i + kR_j$.

The addition of k times the j th column to the i th column is denoted by $C_i + kC_j$.

If a matrix B is obtained from a matrix A by one or more E-operations, then B is said to be equivalent to A .

Two equivalent matrices A and B are written as $A \sim B$.

1.3. ELEMENTARY MATRICES

The matrix obtained from a unit matrix I by subjecting it to one of the E-operations is called an elementary matrix.

For example, let
$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(i) Operating R_{23} or C_{23} on I , we get the same elementary matrix
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

It is denoted by E_{23} . Thus, the E-matrix obtained by either of the operations R_{ij} or C_{ij} on I is denoted by E_{ij} .

(ii) Operating $5R_2$ or $5C_2$ on I , we get the same elementary matrix
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

It is denoted by $5E_2$. Thus, the E-matrix obtained by either of the operations kR_i or kC_i is denoted by kE_i .

(iii) Operating $R_2 + 4R_3$ on I , we get the elementary matrix
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

It is denoted by $E_{23}(4)$. Thus, the E-matrix obtained by the operation $R_i + kR_j$ is denoted by $E_{ij}(k)$.

(iv) Operating $C_2 + 4C_3$ in I , we get the elementary matrix
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}$$
, which is denoted by $E_{23}(4)$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} = E_{23}(4) \text{ and is, the}$$

$C_i + kC_j$ is denoted by $E_{ij}(k)$.

1.4. THE FOLLOWING THEOREMS HOLD GOOD

(a) Any E-row operation on the pre-factor.

If the E-row operation is denoted by $R_i + kR_j$.

(b) Any E-column operation on the post-factor.

If the E-column operation is denoted by $C_i + kC_j$.

(c) Every E-row operation on the pre-factor and every E-column operation on the post-factor are equivalent to the same E-matrix.

Thus the effect of E-row operation on the pre-factor is the same as the effect of E-column operation on the post-factor.

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(d) Every E-column operation on the pre-factor and every E-row operation on the post-factor are equivalent to the same E-matrix.

Thus, the effect of E-column operation on the pre-factor is the same as the effect of E-row operation on the post-factor.

The effect of E-column operation on the pre-factor is the same as the effect of E-row operation on the post-factor.

The effect of E-column operation on the pre-factor is the same as the effect of E-row operation on the post-factor.

1.5. INVERSE OF MATRIX

The elementary row transformation which converts the unit matrix into the inverse of a non-singular square matrix is called the inverse of the matrix.

Let A be a non-singular square matrix of order n .

Apply suitable E-row operations on A to convert it into the unit matrix I .

Simultaneously, apply the same E-row operations on the unit matrix I to convert it into the inverse of A .

so that $I = B$.

Post-multiplying by A^{-1} , we get $IA^{-1} = B A^{-1}$.

$IA^{-1} = B A^{-1}$.

$B A^{-1} = I$.

$B = A$.

Note. In practice, to find the inverse of a matrix, the operations are performed on both the matrix and the unit matrix simultaneously.

1.6. WORKING RULES FOR FINDING THE INVERSE OF A MATRIX OF ORDER 4×4

(i) If in the first column, the first element is not 1, then first interchange the first row with the row in which the first element is 1.

(ii) Operate R_1 on R_2, R_3, R_4 to make the first column as $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$= E_{23}(4)$ and is, therefore, denoted by $E'_{23}(4)$. Thus, the E-matrix obtained by the operation $C_i + kC_j$ is denoted by $E_{ij}'(k)$.

1.4. THE FOLLOWING THEOREMS ON THE EFFECT OF E-OPERATIONS ON MATRICES HOLD GOOD

(a) Any E-row operation on the product of two matrices is equivalent to the same E-row operation on the pre-factor.

If the E-row operation is denoted by R , then $R(AB) = R(A) \cdot B$

(b) Any E-column operation on the product of two matrices is equivalent to the same E-column operation on the post-factor.

If the E-column operation is denoted by C , then $C(AB) = A \cdot C(B)$

(c) Every E-row operation on a matrix is equivalent to pre-multiplication by the corresponding E-matrix.

Thus the effect of E-row operation R_{ij} on $A = E_{ij} \cdot A$

The effect of E-row operation kR_i on $A = kE_i \cdot A$

The effect of E-row operation $R_i + kR_j$ on $A = E_{ij}(k) \cdot A$

(d) Every E-column operation on a matrix is equivalent to post-multiplication by the corresponding E-matrix.

Thus, the effect of E-column operation C_{ij} on $A = A \cdot E_{ij}$

The effect of E-column operation kC_i on $A = A \cdot (kE_i)$

The effect of E-column operation $C_i + kC_j$ on $A = A \cdot E_{ij}'(k)$.

1.5. INVERSE OF MATRIX BY E-OPERATIONS (GAUSS-JORDAN METHOD)

The elementary row transformations which reduce a square matrix A to the unit matrix, when applied to the unit matrix, gives the inverse matrix A^{-1} .

Let A be a non-singular square matrix. Then $A = IA$

Apply suitable E-row operations to A on the left hand side so that A is reduced to I .

Simultaneously, apply the same E-row operations to the pre-factor I on right hand side. Let I reduce to B ,

so that

$$I = BA$$

Post-multiplying by A^{-1} , we get

$$IA^{-1} = BAA^{-1} \Rightarrow A^{-1} = B(AA^{-1}) = BI = B$$

$$\therefore B = A^{-1}$$

Note. In practice, to find the inverse of A by E-row operations, we write A and I side by side and the same operations are performed on both. As soon as A is reduced to I , I will reduce to A^{-1} .

1.6. WORKING RULE TO REDUCE A SQUARE MATRIX TO A UNIT MATRIX I BY ELEMENTARY TRANSFORMATIONS (For Convenience we can consider a Matrix A of order 4×4)

(i) If in the first column, the principle element (i.e., a_{11}) is not 'one' but 'one' is present somewhere else in the first column then first of all make 'one' as principal element.

(ii) Operate R_1 on R_2, R_3, R_4 to make elements of C_1 all zero except first element.

(iii) Then operate R_2 on R_3, R_4 to make elements of C_2 all zero except first and second elements. Similarly operate R_3 on R_4 to make elements of C_3 all zero except 1st, 2nd and 3rd.

(iv) Reduce each diagonal element to element 'one'.

Then reverse process starts :

(v) Operate R_4 on R_1, R_2, R_3 to make all elements of C_4 zero except last.

(vi) Then operate R_3 on R_1 and R_2 to make all elements of C_3 zero except last but one. Similarly operate R_2 on R_1 to make all elements of C_2 zero except last but second and the matrix is reduced to unit matrix.

Note. We can apply the above rule to any square matrix of any order.

ILLUSTRATIVE EXAMPLES

Example 1. Reduce the following matrix to upper triangular form : $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 1 & 2 \end{bmatrix}$

Sol. (Upper triangular matrix) If in a square matrix, all the elements below the principal diagonal are zero, the matrix is called upper triangular.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & -5 & -7 \end{bmatrix} \text{ by operations } \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \text{ by operation } R_3 + 5R_2$$

which is the upper triangular form of the given matrix.

Example 2. Transform $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix}$ in to a unit matrix

Sol. $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & -2 & 4 \\ 0 & -1 & -5 \end{bmatrix} \text{ by operations } R_2 - 2R_1 ; R_3 - 3R_1$

$$\sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & -2 \\ 0 & -1 & -5 \end{bmatrix} \text{ by } R_2 \left(\frac{-1}{2} \right)$$

$$\sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & -7 \end{bmatrix} \text{ by } R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & -7 \end{bmatrix} \text{ by } R_1 - 3R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \text{ by } R_3 \left(-\frac{1}{7} \right)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ by } R_1 - 9R_3 \text{ and } R_2 + 2R_3 \text{ which is unit matrix.}$$

Example 3. Use Gauss-Jordan method to find inverse of the following matrices :

(i) $\begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$ (Mangalore, 1997) (ii) $\begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

Sol. (i) Let

$$A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

Consider $A = IA$ i.e.,

$$\begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Operate R_{13} ,

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 8 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

Operate $R_2 - 2R_1, R_3 - 8R_1$;

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -1 \\ 0 & -12 & -5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ 1 & 0 & -8 \end{bmatrix} A$$

Operate $R_3 - 4R_2$;

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -1 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ 1 & -4 & 0 \end{bmatrix} A$$

Operate $R_1 + R_3, R_2 - R_3$;

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 1 \\ -1 & 5 & -2 \\ 1 & -4 & 0 \end{bmatrix} A$$

Operate $\left(-\frac{1}{3}R_2\right)$ and $(-R_3)$;

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 1 \\ \frac{1}{3} & -\frac{5}{3} & \frac{2}{3} \\ -1 & 4 & 0 \end{bmatrix} A$$

Operate $R_1 - 2R_2$;

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & -\frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{5}{3} & \frac{2}{3} \\ -1 & 4 & 0 \end{bmatrix} A$$

$\therefore$

$I = BA$, where

$$B = A^{-1} = \begin{bmatrix} \frac{1}{3} & -\frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{5}{3} & \frac{2}{3} \\ -1 & 4 & 0 \end{bmatrix}$$

(ii) Let $A = \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$

Operate R_{13} then R_{23} : $\begin{bmatrix} -1 & 4 & 1 & 2 \\ 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$

Operate $R_2 + 2R_1, R_3 + 5R_1$: $\begin{bmatrix} -1 & 4 & 1 & 2 \\ 0 & 2 & 0 & 1 \\ 0 & 7 & 1 & 3 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$

Operate $R_1(-1), R_2\left(\frac{1}{2}\right)$: $\begin{bmatrix} 1 & -4 & -1 & -2 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 7 & 1 & 3 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 & 0 \\ \frac{1}{2} & 0 & 1 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$

Operate $R_3 - 7R_2, R_4 - R_2$: $\begin{bmatrix} 1 & -4 & -1 & -2 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 & 0 \\ \frac{1}{2} & 0 & 1 & 0 \\ -\frac{7}{2} & 1 & -2 & 0 \\ -\frac{1}{2} & 0 & -1 & 1 \end{bmatrix} A$

Operate $R_4(2)$: $\begin{bmatrix} 1 & -4 & -1 & -2 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 & 0 \\ \frac{1}{2} & 0 & 1 & 0 \\ -\frac{7}{2} & 1 & -2 & 0 \\ -1 & 0 & -2 & 2 \end{bmatrix} A$

Operate $R_1 + 2R_4, R_2 - \frac{1}{2}R_4, R_3 + \frac{1}{2}R_4$:

$$\begin{bmatrix} 1 & -4 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & -5 & 4 \\ 1 & 0 & 2 & -1 \\ -4 & 1 & -3 & 1 \\ -1 & 0 & -2 & 2 \end{bmatrix} A$$

Operate $R_1 + R_3$: $\begin{bmatrix} 1 & -4 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -6 & 1 & -8 & 5 \\ 1 & 0 & 2 & -1 \\ -4 & 1 & -3 & 1 \\ -1 & 0 & -2 & 2 \end{bmatrix} A$

Operate $R_1 + 4R_2$;

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 & 1 \\ 1 & 0 & 2 & -1 \\ -4 & 1 & -3 & 1 \\ -1 & 0 & -2 & 2 \end{bmatrix} A$$

$\therefore$

$$I = BA \text{ and } B = A^{-1} = \begin{bmatrix} -2 & 1 & 0 & 1 \\ 1 & 0 & 2 & -1 \\ -4 & 1 & -3 & 1 \\ -1 & 0 & -2 & 2 \end{bmatrix}$$

1.7. NORMAL FORM OF A MATRIX

Any non-zero matrix $A_{m \times n}$ can be reduced to anyone of the following forms by performing elementary (row, column or both) transformations :

$$(i) I_r \quad (ii) [I_r \ 0] \quad (iii) \begin{bmatrix} I_r \\ 0 \end{bmatrix} \quad (iv) \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

All those forms are known as Normal forms of the matrix

Note. (i) The form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ is called First Canonical Form of A

(ii) The number 'r' so obtained is the rank of A (See art 1.9)

[$\because$ rth order minor $\neq 0$ and each $(r+1)$ the order minor = 0]

1.8. FOR ANY MATRIX A OF ORDER $m \times n$, FIND TWO SQUARE MATRICES P AND Q OF ORDERS m AND n RESPECTIVELY SUCH THAT PAQ IS IN THE NORMAL

FORM $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

Method: Write $A = IAI$

Reduce the matrix A on the L.H.S. to normal form by performing elementary row and column transformations.

Every row transformation on A must be accompanied by the same row transformation on the pre-factor on R.H.S.

Every column transformation on A must be accompanied by the same column transformation on the post-factor on R.H.S.

Hence $A = IAI$ will transform to $I = PAQ$

Note. $A^{-1} = QP \because P(AQ) = I \therefore AQ = P^{-1}$ or $(AQ)^{-1} = P$ i.e., $Q^{-1}A^{-1} = P \therefore A^{-1} = QP$.

1.9. RANK OF A MATRIX

(P.T.U., May 2007, May 2008)

Let A be any $m \times n$ matrix. It has square sub-matrices of different orders. The determinants of these square sub-matrices are called minors of A. If all minors of order $(r+1)$ are zero but there is at least one non-zero minor of order r, then r is called the rank of A. Symbolically, rank of A = r is written as $\rho(A) = r$.

From the definition of the rank of a matrix A, it follows that :

(i) If A is a null matrix, then $\rho(A) = 0$

[$\because$ every minor of A has zero value.]

(ii) If A is not a null matrix, then $\rho(A) \geq 1$.

(iii) If A is non-singular $n \times n$ matrix, then $\rho(A) = n$

[$\because |A| \neq 0$ is largest minor of A .]

If I_n is the $n \times n$ unit matrix, then $|I_n| = 1 \neq 0 \Rightarrow \rho(I_n) = n$.

(iv) If A is an $m \times n$ matrix, then $\rho(A) \leq \text{minimum of } m \text{ and } n$.

(v) If all minors of order r are equal to zero, then $\rho(A) < r$.

To determine the rank of a matrix A , we adopt the following different methods :

1.10. WORKING RULE TO DETERMINE THE RANK OF A MATRIX

Method-I : (i) Start with the highest order minor (or minors) of A . Let their order be r . If anyone of them is non-zero, then $\rho(A) = r$.

If all of them are zero, start with minors of next lower order ($r-1$) and so on till you get a non-zero minor. The order of that minor is the rank of A .

This method usually involves a lot of computational work since we have to evaluate several determinants.

Method II : Reduce the matrix to triangular matrix by elementary transformations then number of non-zero rows of triangular matrix is equal to rank of the matrix.

Method III : Reduce the matrix to the normal form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ by performing elementary transformations (row and column both) then r is the rank of the matrix.

1.11. PROPERTIES OF THE RANK OF A MATRIX

(i) Elementary transformations of a matrix do not alter the rank of the matrix.

(ii) $\rho(A') = \rho(A)$; $\rho(A^T) = \rho(A)$

(iii) $\rho(A)$ = number of non-zero rows in upper triangular form of the matrix A .

Example 4. If A is a non-zero column matrix and B is a non-zero row matrix then $\rho(AB) = 1$.

(P.T.U., Dec. 2000)

Sol. Let A be a non-zero column matrix

Let $A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}_{m \times 1}$ and B be a non-zero row matrix

Let $B = [b_1 \ b_2 \ \dots \ b_n]_{1 \times n}$ where
at least one of a 's and at least one of b 's is non-zero

Now AB will be a matrix of order $m \times n$

$$\therefore AB = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} [b_1 \ b_2 \ \dots \ b_n] = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \dots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \dots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_m b_1 & a_m b_2 & \dots & a_m b_n \end{bmatrix}$$

AB has at least one element non-zero and all minor of order ≥ 2 are zero all lines are identical

$\therefore \rho(AB) = 1$.

Example 5. Find the rank of the following matrices

(i) $A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{bmatrix}$ (P.T.U., Dec. 2004)

(ii) $\begin{bmatrix} 1 & 3 & 4 & 5 \\ 1 & 2 & 6 & 7 \\ 1 & 5 & 0 & 10 \end{bmatrix}$

(iii) diag. matrix $[-1 \ 0 \ 1 \ 0 \ 0 \ 4]$.

Sol. (i) Let $A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{bmatrix}$

A is of order $3 \times 3 \quad \therefore \rho(A) \leq 3$

Reduce the matrix to triangular form operate $R_2 - 2R_1, R_3 - 3R_1$

$$A \sim \begin{bmatrix} 1 & 4 & 5 \\ 0 & -2 & -2 \\ 0 & -5 & 7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & -5 & 7 \end{bmatrix} \text{ by } R_2 \left(-\frac{1}{2} \right)$$

$$\sim \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 12 \end{bmatrix} \text{ by } R_3 + 5R_2$$

Now, minor of order 3 = $\begin{vmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 12 \end{vmatrix} = 12 \neq 0$

$\therefore \rho(A) = 3$

(ii) Let $A = \begin{bmatrix} 1 & 3 & 4 & 5 \\ 1 & 2 & 6 & 7 \\ 1 & 5 & 0 & 10 \end{bmatrix}$ A is of order 3×4

$\therefore \rho(A) \leq 3$

Operate $R_2 - R_1, R_3 - R_1 \sim \begin{bmatrix} 1 & 3 & 4 & 5 \\ 0 & -1 & 2 & 2 \\ 0 & 2 & -4 & 5 \end{bmatrix}$

Operate $R_3 + 2R_2 \sim \begin{bmatrix} 1 & 3 & 4 & 5 \\ 0 & -1 & 2 & 2 \\ 0 & 0 & 0 & 9 \end{bmatrix}$

Minor of order 3 i.e., $\begin{vmatrix} 3 & 4 & 5 \\ -1 & 2 & 2 \\ 0 & 0 & 9 \end{vmatrix} \neq 0$

$\therefore \rho(A) = 3$

Matrices

(iii) Let $A = \text{diag. matrix } [-1 \ 0 \ 1 \ 0 \ 0 \ 4]$

$$= \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

it is a square matrix of order $6 \times 6 \therefore \rho(A)$

Also, it is a diagonal matrix so triangular matrix $\therefore \rho(A) = \text{Number of non zero rows of triangular matrix}$
hence $\rho(A) = 3$.

Example 6. Reduce the following matrices to normal form and find their ranks.

(i) $\begin{bmatrix} 2 & 2 & 2 \\ 1 & 2 & 1 \\ 3 & 4 & 3 \end{bmatrix}$ (P.T.U., May 2002)

(ii) $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$

Sol. (i) Let $A = \begin{bmatrix} 2 & 2 & 2 \\ 1 & 2 & 1 \\ 3 & 4 & 3 \end{bmatrix}$

Operate $R_1 \left(\frac{1}{2} \right); \sim \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 3 & 4 & 3 \end{bmatrix}$

Operate $R_2 - R_1, R_3 - 3R_1; \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Operate $R_3 - R_2; \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Operate $R_1 - R_2; \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Operate $C_3 - C_1; \sim \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] = \begin{bmatrix} I_{2 \times 2} & O_{2 \times 1} \\ O_{1 \times 2} & O_{1 \times 1} \end{bmatrix}$

which is the required normal form and $\rho(A) = 2$.

$$(ii) \text{ Let } A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$\text{Operate } R_{12}; \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$\text{Operate } R_2 - 2R_1, R_3 - 3R_1, R_4 - 6R_1; \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

$$\text{Operate } C_2 + C_1, C_3 + 2C_1, C_4 + 4C_1; \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

[Now to change 5 to 1, instead of operating by $R_2 \left(\frac{1}{5} \right)$, operate $R_2 - R_3$ and similarly to change 9 (in 2nd column) to 1 operate $R_4 - 2R_3$]

$$\text{Operate } R_2 - R_3, R_4 - 2R_3; \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 4 & 9 & 10 \\ 0 & 1 & -6 & -3 \end{bmatrix}$$

$$\text{Operate } R_3 - 4R_2, R_4 - R_2; \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Operate } C_3 + 6C_2, C_4 + 3C_2; \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Operate } C_3 \left(\frac{1}{33} \right), C_4 \left(\frac{1}{22} \right); \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Operate } C_4 - C_3; \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cc} I_{3 \times 3} & O_{3 \times 1} \\ O_{1 \times 3} & O_{1 \times 1} \end{array} \right] = \left[\begin{array}{cc} I_3 & 0 \\ 0 & 0 \end{array} \right]$$

which is the required normal form and rank of $A = 3$.

Example 7. Reduce the following matrix to normal form and hence find its rank

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ -2 & 2 & 8 & 0 \end{array} \right]$$

(P.T.U., May 2004)

$$\text{Sol. Let } A = \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ -2 & 2 & 8 & 0 \end{array} \right]$$

$$\text{Operate } R_3 - R_1, R_4 + 2R_1; \sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & 2 & 12 & 2 \end{array} \right]$$

$$\text{Operate } C_3 - 2C_1, C_4 - C_1; \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & 2 & 12 & 2 \end{array} \right]$$

$$\text{Operate } R_3 + R_2, R_4 - 2R_2; \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 16 & 0 \end{array} \right]$$

$$\text{Operate } C_3 + 2C_2, C_4 - C_2; \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 16 & 0 \end{array} \right]$$

$$\text{Operate } R_{34}; \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\text{Operate } R_3 \left(\frac{1}{16} \right); \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right] = \begin{bmatrix} I_3 & O_{3 \times 1} \\ O_{1 \times 3} & O_{1 \times 1} \end{bmatrix}$$

$$= \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \text{ which is the required normal form and } \rho(A) = 3.$$

Example 8. Find non singular matrices P and Q such that PAQ is in the normal form for the matrix

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}. \text{ Also find } A^{-1} \text{ (if it exists)}$$

(P.T.U., Dec. 2003)

Sol. Consider $A = IAI$

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 1 & 1 & 1 & 0 \\ 3 & 1 & 1 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] A \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Operate $R_2 - R_1, R_3 - 3R_1$ (Subjecting prefactor the same operations)

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 4 & 4 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \end{array} \right] A \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Operate $C_2 + C_1, C_3 + C_1$ (Subjecting post factor the same operations)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 4 & 4 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \end{array} \right] A \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Operate $R_3 - 2R_2$ (Same operation on prefactor on R.H.S.)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & -2 & 1 & 0 \end{array} \right] A \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Operate $C_3 - C_2$ (Same operation on post factor on R.H.S.)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & -2 & 1 & 0 \end{array} \right] A \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Operating $R_2 \left(\frac{1}{2} \right)$ (Same operation on prefactor)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ -1 & -2 & 1 & 0 \end{array} \right] A \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} = PAQ, \text{ where } P = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -1 & -2 & 1 \end{bmatrix} \text{ and } Q = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

As $\rho(A) = 2 \therefore A$ is a singular matrix and A^{-1} does not exist.

TEST YOUR KNOWLEDGE

1. Reduce to triangular form $\begin{bmatrix} 3 & -4 & -5 \\ -9 & 1 & 4 \\ -5 & 3 & 1 \end{bmatrix}$

2. Use Gauss Jordan method to find the inverse of the following

(i) $\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$

(ii) $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$

(iii) $\begin{bmatrix} 2 & 4 & 3 & 2 \\ 3 & 6 & 5 & 2 \\ 2 & 5 & 2 & -3 \\ 4 & 5 & 14 & 14 \end{bmatrix}$

(iv) $\begin{bmatrix} 2 & 1 & -1 & 2 \\ 1 & 3 & 2 & -3 \\ -1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{bmatrix}$

3. Find rank of the following matrices

(i) $\begin{bmatrix} 2 & -1 & 0 & 5 \\ 0 & 3 & 1 & 4 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$

(iii) $\begin{bmatrix} 2 & 4 & 3 & -2 \\ -3 & -2 & -1 & 4 \\ 6 & -1 & 7 & 2 \end{bmatrix}$

(iv) $\begin{bmatrix} 2 & 5 & 3 & 3 \\ 2 & 3 & 3 & 4 \\ 3 & 6 & 3 & 2 \\ 4 & 12 & 0 & 8 \end{bmatrix}$

(v) $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix}$ (P.T.U., June 2003)

4. Reduce the following matrices to normal form and hence find their ranks

(i) $\begin{bmatrix} 2 & 2 & 2 \\ 1 & 2 & 1 \\ 3 & 4 & 3 \end{bmatrix}$

(ii) $\begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 0 & 5 & -10 \end{bmatrix}$

(iv) $\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$

(v) $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

(Karnataka, 1993; Punjab)

5. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$; find two non-singular matrices P and Q such that $PAQ = I$. Hence find A^{-1} .

[Hint: $\rho(A) = 3 \therefore A^{-1}$ exists = QP]

6. For a matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$ find non-singular matrices P and Q such that PAQ is in the normal form. Also find A^{-1} (if it exists).

ANSWERS

1. $\begin{bmatrix} 3 & 4 & -5 \\ 0 & 13 & -11 \\ 0 & 0 & 1 \end{bmatrix}; \begin{bmatrix} 3 & 0 & 0 \\ -9 & 13 & 0 \\ -15 & 29 & 11 \end{bmatrix}$
Upper triangular Lower triangular

2. (i) $\begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$

(ii) $\begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}$

(iii) $\frac{1}{25} \begin{bmatrix} -23 & 29 & -64 & -18 \\ 10 & -12 & 26 & 7 \\ 1 & -2 & 6 & 2 \\ 2 & -2 & 3 & 1 \end{bmatrix}$

(iv) $\frac{1}{18} \begin{bmatrix} 2 & 5 & -7 & 1 \\ 5 & -1 & 5 & -2 \\ -7 & 5 & 11 & 10 \\ -1 & -2 & 10 & 5 \end{bmatrix}$

3. (i) 2

(ii) 2

(iii) 3

(iv) 4

(v) 3

4. (i) $\begin{bmatrix} I_2 & O_{2 \times 1} \\ O_{1 \times 2} & O_{1 \times 1} \end{bmatrix}; \text{rank} = 2$

(ii) $\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] = \left[\begin{array}{cc} I_2 & 0 \\ 0 & 0 \end{array} \right]; \rho = 2$

(iii) $[I_3 \times O_{3 \times 1}]; \rho = 3$

(iv) $\begin{bmatrix} I_2 & O_{2 \times 2} \\ O_{2 \times 2} & O_{2 \times 2} \end{bmatrix}; \rho = 2$

5. $P = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & -3 \end{bmatrix}; Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}; A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$

6. $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}; Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}; A^{-1}$ does not exist as $\rho(A) = 2$.

1.12. CONSISTENCY AND SOLUTION OF ALGEBRAIC EQUATIONS

Proof. Consider the system of equations $\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$ (3 equations in 3 unknowns)

$$\begin{bmatrix} a_1x + b_1y + c_1z \\ a_2x + b_2y + c_2z \\ a_3x + b_3y + c_3z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \text{ or } \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

or

$$AX = B$$

where $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ is called the coefficient matrix, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is the column matrix of unknowns

$B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$ is the column of constants.

If $d_1 = d_2 = d_3 = 0$, then $B = O$ and the matrix equation $AX = B$ reduce to $AX = O$.

Such a system of equation is called a system of **homogeneous linear equation**.

If at least one of d_1, d_2, d_3 is non-zero, then $B \neq O$.

Such a system of equation is called a system of **non-homogeneous linear equation**.

Solving the matrix equation $AX = B$ means finding X , i.e., finding a column matrix $\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$ such that

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}. \text{ Then } x = \alpha, y = \beta, z = \gamma.$$

The matrix equation $AX = B$ need not always have a solution. It may have no solution or a unique solution or an infinite-number of solution.

(a) **Consistent Equations** : A system of equations having **one or more solutions** is called a consistent system of equations.

(b) **Inconsistent Equations** : A system of equations having **no solutions** is called an inconsistent system of equations.

(c) **State the conditions in terms of rank of the coefficient matrix and rank of the augmented matrix for a unique solution; no solution; infinite number of solutions of a system of linear equations.**

(P.T.U., May 2002)

For a system of non-homogeneous linear equation $AX = B$.

(i) if $\rho[A : B] \neq \rho(A)$, the system is inconsistent.

(ii) if $\rho[A : B] = \rho(A) = \text{number of the unknowns}$, the system has a unique solution.

(iii) if $\rho[A : B] = \rho(A) < \text{number of unknowns}$, the system has an infinite number of solutions.

The matrix $[A : B]$ in which the elements of A and B are written side by side is called the **augmented matrix**.

For a system of homogeneous linear equations $AX = O$.

(i) $X = O$ is always a solution. This solution in which each unknown has the value zero is called the **Trivial Solution**. Thus a homogeneous system is always consistent.

(ii) if $\rho(A) = \text{number of unknown}$, the system has only the trivial solution.

(iii) if $\rho(A) < \text{number of unknown}$, the system has an infinite number of non-trivial solutions.

1.13. IF A IS A NON-SINGULAR MATRIX, THEN THE MATRIX EQUATION $AX = B$ HAS A UNIQUE SOLUTION

The given equation is $AX = B$

...(1)

$\therefore A$ is a non-singular matrix, $\therefore A^{-1}$ exists.

Pre-multiplying both sides of (1) by A^{-1} , we get

$$A^{-1}(AX) = A^{-1}B \quad \text{or} \quad (A^{-1}A)X = A^{-1}B$$

or

$$IX = A^{-1}B \quad \text{or} \quad X = A^{-1}B$$

which is the required unique solution (since A^{-1} is unique).

Another Method

Write the augmented matrix $[A : B]$. By E-row operations on A and B, reduce A to a diagonal matrix thus getting

$$[A : B] \sim \begin{bmatrix} p_1 & 0 & 0 & : & q_1 \\ 0 & p_2 & 0 & : & q_2 \\ 0 & 0 & p_3 & : & q_3 \end{bmatrix}$$

Then

$$p_1x = q_1, \quad p_2y = q_2, \quad p_3z = q_3.$$

ILLUSTRATIVE EXAMPLES

Example 1. Solve the system of equations :

$$5x + 3y + 7z = 4, \quad 3x + 26y + 2z = 9, \quad 7x + 2y + 11z = 5$$

with the help of matrix inversion.

(P.T.U., May 2007; Dec. 2004, ; Nagpur, 1997; Mysore, 1994; Delhi, 1994;

Bhopal 1991; Rewa, 1990; Kerala, 1989 S, 95)

Sol. In matrix notation, the given system of equations can be written as

$$AX = B$$

...(1)

where $A = \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 11 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$

Let $A = \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 11 \end{bmatrix}$

$$|A| = \begin{vmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 11 \end{vmatrix} = 5 \begin{vmatrix} 26 & 2 \\ 2 & 11 \end{vmatrix} - 3 \begin{vmatrix} 3 & 2 \\ 7 & 11 \end{vmatrix} + 7 \begin{vmatrix} 3 & 26 \\ 7 & 2 \end{vmatrix}$$

$$= 5(286 - 4) - 3(33 - 14) + 7(6 - 182) = 1410 - 57 - 1232 = 121 \neq 0$$

$\Rightarrow A$ is non-singular $\therefore A^{-1}$ exists and the unique solution of (1) is

$$X = A^{-1}B$$

...(2)

$$A_1 = \begin{vmatrix} 26 & 2 \\ 2 & 11 \end{vmatrix} = 282, A_2 = -\begin{vmatrix} 3 & 7 \\ 2 & 11 \end{vmatrix} = -19, A_3 = \begin{vmatrix} 3 & 26 \\ 7 & 2 \end{vmatrix} = -176$$

$$B_1 = \begin{vmatrix} 3 & 2 \\ 7 & 11 \end{vmatrix} = -19, B_2 = \begin{vmatrix} 5 & 7 \\ 7 & 11 \end{vmatrix} = 6, B_3 = -\begin{vmatrix} 5 & 7 \\ 3 & 2 \end{vmatrix} = 11$$

Matrices

$$C_1 = \begin{vmatrix} 3 & 26 \\ 7 & 2 \end{vmatrix} = -176, C_2 = -\begin{vmatrix} 5 & 3 \\ 7 & 2 \end{vmatrix} = 11, C_3 = \begin{vmatrix} 5 & 3 \\ 3 & 26 \end{vmatrix} = 121$$

$$\therefore \text{adj } A = \text{transpose of } \begin{bmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \begin{bmatrix} 282 & -19 & -176 \\ -19 & 6 & 11 \\ -176 & 11 & 121 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{121} \begin{bmatrix} 282 & -19 & -176 \\ -19 & 6 & 11 \\ -176 & 11 & 121 \end{bmatrix}$$

$$\text{From (2), } X = A^{-1}B = \frac{1}{121} \begin{bmatrix} 282 & -19 & -176 \\ -19 & 6 & 11 \\ -176 & 11 & 121 \end{bmatrix} \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix} = \frac{1}{121} \begin{bmatrix} 282(4) - 19(9) - 176(5) \\ -19(4) + 6(9) + 11(5) \\ -176(4) + 11(9) + 121(5) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{121} \begin{bmatrix} 77 \\ 33 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{77}{121} \\ \frac{33}{121} \\ \frac{0}{121} \end{bmatrix} = \begin{bmatrix} \frac{7}{11} \\ \frac{3}{11} \\ 0 \end{bmatrix}$$

Hence $x = \frac{7}{11}, y = \frac{3}{11}, z = 0$.

Example 2. Solve the system of equations:

$$\begin{aligned} 2x_1 + x_2 + 2x_3 + x_4 &= 6, & 6x_1 - 6x_2 + 6x_3 + 12x_4 &= 36 \\ 4x_1 + 3x_2 + 3x_3 - 3x_4 &= -1, & 2x_1 + 2x_2 - x_3 + x_4 &= 10 \end{aligned}$$

Sol. In matrix notation, the given system of equations can be written as $AX = B$

$$\text{where } A = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 6 & -6 & 6 & 12 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, B = \begin{bmatrix} 6 \\ 36 \\ -1 \\ 10 \end{bmatrix}$$

[Note: In this question there is no restriction that solution must be obtained by matrix inversion; it is very difficult to evaluate determinants of order four and three]

$\therefore$ it is advisable to apply method of Augmented matrix]

Augmented matrix $[A : B]$

$$= \begin{bmatrix} 2 & 1 & 2 & 1 & : & 6 \\ 6 & -6 & 6 & 12 & : & 36 \\ 4 & 3 & 3 & -3 & : & -1 \\ 2 & 2 & -1 & 1 & : & 10 \end{bmatrix} \text{operating } R_2 - 3R_1, R_3 - 2R_1, R_4 - R_1$$

$$= \begin{bmatrix} 2 & 1 & 2 & 1 & : & 6 \\ 0 & -9 & 0 & 9 & : & 18 \\ 0 & 1 & -1 & -5 & : & -13 \\ 0 & 1 & -3 & 0 & : & 4 \end{bmatrix} \text{operating } -\frac{1}{9}R_2$$

$$\sim \begin{bmatrix} 2 & 1 & 2 & 1 & : & 6 \\ 0 & 1 & 0 & -1 & : & -2 \\ 0 & 1 & -1 & -5 & : & -13 \\ 0 & 1 & -3 & 0 & : & 4 \end{bmatrix} \text{operating } R_1 - R_2, R_3 - R_2, R_4 - R_2$$

$$\sim \begin{bmatrix} 2 & 0 & 2 & 2 & : & 8 \\ 0 & 1 & 0 & -1 & : & -2 \\ 0 & 0 & -1 & -4 & : & -11 \\ 0 & 0 & -3 & 1 & : & 6 \end{bmatrix} \text{operating } R_4 - 3R_3, \frac{1}{2}R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 & : & 4 \\ 0 & 1 & 0 & -1 & : & -2 \\ 0 & 0 & -1 & -4 & : & -11 \\ 0 & 0 & 0 & 13 & : & 39 \end{bmatrix} \text{operating } R_1 + R_3, \frac{1}{13}R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -3 & : & 7 \\ 0 & 1 & 0 & -1 & : & -2 \\ 0 & 0 & -1 & -4 & : & -11 \\ 0 & 0 & 0 & 1 & : & 3 \end{bmatrix} \text{operating } R_1 + 3R_4, R_2 + R_4, R_3 + 4R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & 0 & : & 1 \\ 0 & 0 & -1 & 0 & : & 1 \\ 0 & 0 & 0 & 1 & : & 3 \end{bmatrix} \text{operating } (-1)R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & 0 & : & 1 \\ 0 & 0 & 1 & 0 & : & -1 \\ 0 & 0 & 0 & 1 & : & 3 \end{bmatrix}$$

Since $\rho(A) = \rho(A : B) = 4 \therefore$ systems of equations have a unique solution.

Hence, $x_1 = 2, x_2 = 1, x_3 = -1, x_4 = 3$.

Example 3. Using matrix method, show that the equations :

$$3x + 3y + 2z = 1, \quad x + 2y = 4, \quad 10y + 3z = -2, \quad 2x - 3y - z = 5$$

are consistent and hence obtain the solution for x, y and z .

Sol. In matrix notation, the given system of equations can be written as

$$AX = B$$

where $A = \begin{bmatrix} 3 & 3 & 2 \\ 1 & 2 & 0 \\ 0 & 10 & 3 \\ 2 & -3 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 1 \\ 4 \\ -2 \\ 5 \end{bmatrix}$

Augmented matrix $[A : B]$

$$= \begin{bmatrix} 3 & 3 & 2 & : & 1 \\ 1 & 2 & 0 & : & 4 \\ 0 & 10 & 3 & : & -2 \\ 2 & -3 & -1 & : & 5 \end{bmatrix} \text{operating } R_{12}$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 3 & 3 & 2 & : & 1 \\ 0 & 10 & 3 & : & -2 \\ 2 & -3 & -1 & : & 5 \end{bmatrix} \text{operating } R_2 - 3R_1, R_4 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 0 & -3 & 2 & : & -11 \\ 0 & 10 & 3 & : & -2 \\ 0 & -7 & -1 & : & -3 \end{bmatrix} \text{operating } R_3 + 3R_2, R_4 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & : & 4 \\ 0 & -3 & 2 & : & -11 \\ 0 & 1 & 9 & : & -35 \\ 0 & -1 & -5 & : & 19 \end{bmatrix} \text{operating } R_1 - 2R_3, R_2 + 3R_3, R_4 + R_3$$

$$\sim \begin{bmatrix} 1 & 0 & -18 & : & 74 \\ 0 & 0 & 29 & : & -116 \\ 0 & 1 & 9 & : & -35 \\ 0 & 0 & 4 & : & -16 \end{bmatrix} \text{operating } R_{23}, \frac{1}{4}R_4$$

$$\sim \begin{bmatrix} 1 & 0 & -18 & : & 74 \\ 0 & 1 & 9 & : & -35 \\ 0 & 0 & 29 & : & -116 \\ 0 & 0 & 1 & : & -4 \end{bmatrix} \text{operating } R_1 + 18R_4, R_2 - 9R_4, R_3 - 29R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & 1 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 1 & : & -4 \end{bmatrix} \text{operating } R_{34}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & 1 \\ 0 & 0 & 1 & : & -4 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\rho(A) = \rho(A : B) = 3 = \text{number of unknowns}$$

$\Rightarrow$ The given system of equation is consistent and the solution is $x = 2, y = 1, z = -4$.

Example 4. For what values of λ and μ do the system of equations:

$$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$$

have (i) no solution (ii) unique solution (iii) more than one solution ?

(P.T.U., Dec. 2002 ; Panjab, 1990 ; Rewa, 1990 ; A.M.I.E.T.E 2001)

Sol. In matrix notation, the given system of the equations can be written as

$$AX = B$$

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$

Augmented matrix $[A : B]$

$$= \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix} \text{operating } R_2 - R_1, R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & \lambda - 1 & : & \mu - 6 \end{bmatrix}$$

Operating $R_1 - R_2, R_3 - R_2$

$$\sim \begin{bmatrix} 1 & 0 & -1 & : & 2 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda - 3 & : & \mu - 10 \end{bmatrix}$$

Case I. If $\lambda = 3, \mu \neq 10$

$$\rho(A) = 2, \rho(A : B) = 3$$

$$\therefore \rho(A) \neq \rho(A : B)$$

$\therefore$ The system has no solution.

Case II. If $\lambda \neq 3, \mu$ may have any value

$$\rho(A) = \rho(A : B) = 3 = \text{number of the unknowns}$$

$\therefore$ System has unique solution.

Case III. If $\lambda = 3, \mu = 10$

$$\rho(A) = \rho(A : B) = 2 < \text{number of the unknowns}$$

$\therefore$ The system has an infinite number of solution.

Example 5. Solve the equations :

$$x_1 + 3x_2 + 2x_3 = 0, 2x_1 - x_2 + 3x_3 = 0, 3x_1 - 5x_2 + 4x_3 = 0, x_1 + 17x_2 + 4x_3 = 0.$$

Sol. In matrix notation, the given system of equations can be written as

$$AX = 0$$

where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & -1 & 3 \\ 3 & -5 & 4 \\ 1 & 17 & 4 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Operating $R_2 - 2R_1, R_3 - 3R_1, R_4 - R_1$

$$A \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -7 & -1 \\ 0 & -14 & -2 \\ 0 & 14 & 2 \end{bmatrix}$$

Operating $R_3 - 2R_2, R_4 + 2R_2$

$$\sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -7 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Matrices

Operating $R_1 + 2R_2$

$$\sim \begin{bmatrix} 1 & -11 & 0 \\ 0 & -7 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow \rho(A) = 2 < \text{number of unknowns}$

The system has an infinite number of non-trivial solutions given by

$$x_1 - 11x_2 = 0, -7x_2 - x_3 = 0$$

$$x_1 = 11k, x_2 = k, x_3 = -7k, \text{ where } k \text{ is any number.}$$

i.e.,

Different values of k give different solutions.

Example 6. Discuss the consistency of the following system of equations. Find the solution, (P)

$$(i) \quad \begin{aligned} x + y + z &= 4 \\ 2x + 5y - 2z &= 3 \end{aligned} \quad (P)$$

$$(ii) \quad \begin{aligned} 5x + 3y + 7z &= 4 \\ 3x + 26y + 2z &= 9 \\ 7x + 2y + 10z &= 5. \end{aligned} \quad (P)$$

Sol. (i) $x + y + z = 4$
 $2x + 5y - 2z = 3$ which can be written

as

$$AX = B, \text{ where } A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & -2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

consider augmented matrix $[A : B]$

$$= \begin{bmatrix} 1 & 1 & 1 & : & 4 \\ 2 & 5 & -2 & : & 3 \end{bmatrix}$$

Operating $R_2 - 2R_1$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 4 \\ 0 & 3 & -4 & : & -5 \end{bmatrix}$$

Operating $R_2 \left(\frac{1}{3} \right)$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 4 \\ 0 & 1 & -\frac{4}{3} & : & -\frac{5}{3} \end{bmatrix}$$

Operating $R_1 - R_2$

$$\begin{bmatrix} 1 & 0 & \frac{7}{3} & : & \frac{17}{3} \\ 0 & 1 & -\frac{4}{3} & : & -\frac{5}{3} \end{bmatrix}$$

$$\rho(A) = 2; \rho(A : B) = 2$$

$$\rho(A) = \rho(A : B) < \text{number of unknowns}$$

∴ Given system of equations are consistent and have infinite number of solutions given by

$$x + \frac{7}{3}z = \frac{17}{3}$$

$$y - \frac{4}{3}z = -\frac{5}{3}$$

Take $z = k$ we have $x = \frac{17-7k}{3}$, $y = \frac{4k-5}{3}$

Hence solutions is $x = \frac{17-7k}{3}$, $y = \frac{4k-5}{3}$, $z = k$, where k is any arbitrary constant.

(ii) Given equations can be put into the form $AX = B$

where

$$A = \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$$

Consider Augmented matrix

$$[A : B] = \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 3 & 26 & 2 & : & 9 \\ 7 & 2 & 10 & : & 5 \end{bmatrix}$$

Operating $R_2 - \frac{3}{5}R_1, R_3 - \frac{7}{5}R_1$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & \frac{121}{5} & -\frac{11}{5} & : & \frac{33}{5} \\ 0 & -\frac{11}{5} & \frac{1}{5} & : & -\frac{3}{5} \end{bmatrix}$$

Operate $R_3 + \frac{1}{11}R_2$; $\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & \frac{121}{5} & -\frac{11}{5} & : & \frac{33}{5} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$

Operate $R_2 \left(\frac{5}{11}\right) \sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 11 & -1 & : & 3 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$

$$\rho(A) = 2, \rho(A : B) = 2$$

$$\rho(A) = \rho(A : B) = 2 < \text{Number of unknowns}$$

∴ Given equations are consistent and have infinite number of solutions given by

$$5x + 3y + 7z = 4$$

$$11y - z = 3$$

Let

$$z = k; y = \frac{k+3}{11} \therefore x = \frac{4 - \frac{3(k+3)}{11} - 7k}{5}$$

$$= \frac{7-16k}{11}$$

Hence solution is

$$x = \frac{7-16k}{11}, y = \frac{k+3}{11}, z = k.$$

Example 7. For what value of k , the equations $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$ have a solution and solve them completely in each case.
(P.T.U., Dec. 2005, May 2001)

Sol.

$$\begin{aligned} x + y + z &= 1 \\ 2x + y + 4z &= k \\ 4x + y + 10z &= k^2 \end{aligned}$$

which can be put into matrix form

$$AX = B$$

where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 4 \\ 4 & 1 & 10 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 1 \\ k \\ k^2 \end{bmatrix}$$

Consider the Augmented matrix

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 2 & 1 & 4 & : & k \\ 4 & 1 & 10 & : & k^2 \end{bmatrix}$$

Operate $R_2 - 2R_1, R_3 - 4R_1$, we get ~

$$\begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & k-2 \\ 0 & -3 & 6 & : & k^2-4 \end{bmatrix}$$

Operate $R_3 - 3R_2$ ~

$$\begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & k-2 \\ 0 & 0 & 0 & : & k^2-3k+2 \end{bmatrix}$$

$$\rho(A) = 2 < \text{number of unknowns}$$

$\therefore$ System of equations cannot have a unique solution.

These will have an infinite number of solution only if $\rho(A : B) = \rho(A) = 2$ which is only possible if $k^2 - 3k + 2 = 0$ i.e., $k = 1$ or $k = 2$.

when

$$k = 1; \text{ the augmented matrix } = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & -1 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\therefore$ Equations are $x + y + z = 1$
 $-y + 2z = -1$

Let

$$z = \lambda, y = 1 + 2\lambda, x = -3\lambda. \text{ where } \lambda \text{ is an arbitrary constant}$$

when

$$k = 2; \text{ Augmented matrix } = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

Equations are $x + y + z = 1$ and $-y + 2z = 0$.

Take

$$z = \lambda', y = 2\lambda', x = 1 - 3\lambda'$$

Where λ' is any arbitrary constant.

Example 8. Find the values of a and b for which the equations $x + ay + z = 3$; $x + 2y + 2z = b$, $x + 5y + 3z = 9$ are consistent. When will these equations have a unique solution? (P.T.U., Dec. 2003)

Sol. Given equations are

$$x + ay + z = 3$$

$$x + 2y + 2z = b$$

$$x + 5y + 3z = 9.$$

The matrix equation is $AX = B$, where

$$A = \begin{bmatrix} 1 & a & 1 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \end{bmatrix}, B = \begin{bmatrix} 3 \\ b \\ 9 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Augmented matrix is $\left[\begin{array}{ccc|c} 1 & a & 1 & 3 \\ 1 & 2 & 2 & b \\ 1 & 5 & 3 & 9 \end{array} \right]$

Operate $R_2 - R_1, R_3 - R_1 \sim \left[\begin{array}{ccc|c} 1 & a & 1 & 3 \\ 0 & 2-a & 1 & b-3 \\ 0 & 5-a & 2 & 6 \end{array} \right]$

Operate $R_3 - \frac{5-a}{2-a} R_2 \sim \left[\begin{array}{ccc|c} 1 & a & 1 & 3 \\ 0 & 2-a & 1 & b-3 \\ 0 & 0 & 2-\frac{5-a}{2-a} & 6-\frac{5-a}{2-a}(b-3) \end{array} \right]$

or $\left[\begin{array}{ccc|c} 1 & a & 1 & 3 \\ 0 & 2-a & 1 & b-3 \\ 0 & 0 & \frac{-1-a}{2-a} & 6-\frac{5-a}{2-a}(b-3) \end{array} \right]$

Case I. If $\frac{-1-a}{2-a} = 0$ and $6 - \frac{5-a}{2-a}(b-3) = 0$

then $a = -1 \therefore b = 6$ i.e., $a = -1, b = 6$

$\rho(A) = \rho(A:B) = 2 < \text{number of unknowns}$

$\therefore$ Given equations have infinite number of solutions given by

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 3 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Equations are $x - y + z = 3$

$$3y + z = 3$$

Let

$$z = k; y = \frac{3-k}{3} \therefore x = 3 + \frac{3-k}{3} - k = 4 - \frac{4}{3}k$$

$\therefore$

$$x = 4 - \frac{4}{3}k, y = \frac{3-k}{3}, z = k \text{ is the solution.}$$

Matrices

Case II. If $a = -1$, but $b \neq 6$

Then $\rho(A) = 2$, but $\rho(A : B) = 3$

$\rho(A) \neq \rho(A : B) \therefore$ Equations are inconsistent, i.e., having no solution.

Case III. If $a \neq -1$, b can have any value then $\rho(A) = 3 = \rho(A : B)$

$\therefore$ Given equations have a unique solution which is given by the equation.

$$x + ay + z = 3$$

$$(2-a)y + z = b-3$$

$$\frac{-(1+a)}{2-a} z = 6 - \frac{5-a}{2-a} (b-3)$$

$$z = \frac{2-a}{-(1+a)} \left\{ 6 - \frac{(5-a)(b-3)}{2-a} \right\} = \frac{6(2-a) - (5-a)(b-3)}{-(1+a)}$$

$$y = \frac{1}{2-a} \left[(b-3) - \frac{6(2-a) - (5-a)(b-3)}{-(1+a)} \right]$$

$$= \frac{1}{(2-a)(1+a)} [(b-3)(1+a) + 6(2-a) - (5-a)(b-3)]$$

$$y = \frac{2(6-b)}{1+a}; x = \frac{(5-3a)(6-b)}{1+a}$$

TEST YOUR KNOWLEDGE

1. Write the following equations in matrix form $AX = B$ and solve for X by finding A^{-1} .

$$(i) \quad 2x - 2y + z = 1 \quad (ii) \quad 2x_1 - x_2 + x_3 = 4$$

$$x + 2y + 2z = 2 \quad x_1 + x_2 + x_3 = 1$$

$$2x + y - 2z = 7 \quad x_1 - 3x_2 - 2x_3 = 2$$

2. Using the loop current method on a circuit, the following equations were obtained :

$$7i_1 - 4i_2 = 12, -4i_1 + 12i_2 - 6i_3 = 0, 6i_2 + 14i_3 = 0.$$

By matrix method, solve for i_1, i_2 and i_3

3. Solve the following system of equations by matrix method :

$$(i) \quad x + y + z = 8, \quad x - y + 2z = 6, \quad 3x + 5y - 7z = 14$$

$$(ii) \quad x + y + z = 6, \quad x - y + 2z = 5, \quad 3x + y + z = 3$$

$$(iii) \quad x + 2y + 3z = 1, \quad 2x + 3y + 2z = 2, \quad 3x + 3y + 4z = 1,$$

4. Show that the equations $x + 2y - z = 3, 3x - y + 2z = 1, 2x - 2y + 3z = 2, x - y + z = -1$ are consistent.

5. Test for consistency the equations $2x - 3y + 7z = 5, 3x + y - 3z = 13, 2x + 19y - 47z = 32$.

(Bhopal, 1991)

6. Solve the equations $x + 3y + 2z = 0, 2x - y + 3z = 0, 3x - 5y + 4z = 0, x + 17y + 4z = 0$.

7. For what values of a and b do the equations $x + 2y + 3z = 6, x + 3y + 5z = 9, 2x + 5y + az = b$ have
(i) a unique solution (ii) more than one solution?

8. Find the value of k so that the equations $x + y + 3z = 0$, $4x + 3y + kz = 0$, $2x + y + 2z = 0$ have a non-trivial solution.
9. Show that the equations $3x + 4y + 5z = a$, $4x + 5y + 6z = b$, $5x + 6y + 7z = c$ do not have a solution unless $a + c = 2b$.
10. Investigate the value of λ and μ so that the equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + \lambda z = \mu$ have (i) no solution, (ii) a unique solution and (iii) an infinite number of solution. (Kerala, 1990)
11. Determine the value of λ for which the following set of equations may possess non-trivial solution.
 $3x_1 + x_2 - \lambda x_3 = 0$, $4x_1 - 2x_2 - 3x_3 = 0$, $2\lambda x_1 + 4x_2 + \lambda x_3 = 0$. For each permissible value of λ , determine the general solution. (Kerala, 1991)
12. Investigate for consistency of the following equations and if possible find the solutions.
 $4x - 2y + 6z = 8$, $x + y - 3z = -1$, $15x - 3y + 9z = 21$. (Mysore, 1994; Rewa, 1994)
13. Show that if $\lambda \neq -5$, the system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ have a unique solution. If $\lambda = -5$, show that the equations are consistent. Determine the solutions in each case. (Marathwada, 1993)
14. Show that the equations $2x + 6y + 11 = 0$, $6x + 20y - 6z + 3 = 0$, $6y - 18z + 1 = 0$ are not consistent. (P.T.U., Dec. 2003)

ANSWERS

1. (i) $x = 2, y = 1, z = -1$ (ii) $x_1 = 1, x_2 = -1, x_3 = 1$ 2. $i_1 = \frac{396}{175}, i_2 = \frac{24}{25}, i_3 = \frac{72}{175}$
3. (i) $x = 5, y = \frac{5}{3}, z = \frac{4}{3}$ (ii) $x = 1, y = 2, z = 3$ (iii) $x = -\frac{3}{7}, y = \frac{8}{7}, z = -\frac{2}{7}$
4. $x = -1, y = 4, z = 4$ 5. Inconsistent 6. $x = 11k, y = k, z = -7k$, where k is arbitrary
7. (i) $a = 8, b \neq 15$ (ii) $z \neq 8, b$ may have any value (iii) $a = 8, b = 15$
8. $k = 8$ 10. (i) $\lambda = 5, \mu \neq 9$ (ii) $\lambda \neq 5, \mu$ arbitrary (iii) $\lambda = 5, \mu = 9$
11. $\lambda = 1, -9$ for $\lambda = 1$ solution is $x = k = -k, z = 2k$. For $\lambda = -9$ solution is $x = 3k, y = 9k, z = -2k$.
12. Consistent : $x = 1, y = 3k - 2, z = k$, where k is arbitrary
13. $\lambda \neq -5, x = \frac{4}{7}, y = -\frac{9}{7}, z = 0$; $\lambda = -5, x = \frac{4 - 5k}{7}, y = \frac{13k - 9}{7}, z = k$, where k is arbitrary.

1.14. VECTORS

Any ordered n -tuple of numbers is called an n -vector. By an ordered n -tuple, we mean a set consisting of n numbers in which the place of each number is fixed. If $x_1, x_2, \dots, x_n$ be any n numbers then the ordered n -tuple $X = (x_1, x_2, \dots, x_n)$ is called an n -vector. Thus the co-ordinates of a point in space can be represented by a 3-vector. (x, y, z) . Similarly $(1, 0, 2, -1)$ and $(2, 7, 5, -3)$ are 4-vectors. The n numbers $x_1, x_2, \dots, x_n$ are called the components of the n -vector $X = (x_1, x_2, \dots, x_n)$. A vector may be written either as a row vector or as a column vector. If A be a matrix of order $m \times n$, then each row of A will be an n -vector and each column of A will be an m -vector. A vector whose components are all zero is called a zero vector and is denoted by O . Thus $O = (0, 0, 0, \dots, 0)$.

Let $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n)$ be two vectors.

Then $X = Y$ if and only if their corresponding components are equal.

i.e., if $x_i = y_i$ for $i = 1, 2, \dots, n$

If k be a scalar, then $kX = (kx_1, kx_2, \dots, kx_n)$.

1.15. LINEAR DEPENDENCE AND LINEAR INDEPENDENCE OF VECTORS

29

(P.T.U., May 2004, 2006)

A set of r , n -tuple vectors $X_1, X_2, \dots, X_r$ is said to be *linearly dependent* if there exists r scalars (numbers) $k_1, k_2, \dots, k_r$, not all zero, such that

$$k_1 X_1 + k_2 X_2 + \dots + k_r X_r = 0$$

A set of r , n -tuple vectors $X_1, X_2, \dots, X_r$ is said to be *linearly independent* if every relation of the type $k_1 X_1 + k_2 X_2 + \dots + k_r X_r = 0$ implies $k_1 = k_2 = \dots = k_r = 0$

To test the linear dependence of r given vectors, write them as row vectors. Add suitable multiples of one vector to the others so that the resulting $(r-1)$ vector have their first component zero. Choose anyone of these $(r-1)$ vectors and add its multiples to the others so that the resulting $(r-2)$ vectors have their second component zero. In this way continue, reducing the successive components to zero. If the final reduction gives a vector all of whose components are zero, then the original vectors are linearly dependent. However, if the final reduction given a vector all of whose components are not zero, then the original vectors are linearly independent.

Note. If a set of vectors is linearly dependent, then at least one member of the set can be expressed as a linear combination of the remaining vectors.

Example 1. Show that the vectors $x_1 = (1, 2, 4)$, $x_2 = (2, -1, 3)$, $x_3 = (0, 1, 2)$ and $x_4 = (-3, 7, 2)$ are linearly dependent and find the relation between them. (Bhopal, 1991)

Sol. Consider the matrix equation,

$$\begin{aligned} k_1 x_1 + k_2 x_2 + k_3 x_3 + k_4 x_4 &= 0 \\ \text{i.e., } k_1 (1, 2, 4) + k_2 (2, -1, 3) + k_3 (0, 1, 2) + k_4 (-3, 7, 2) &= 0 \\ \text{i.e., } k_1 + 2k_2 + 0 \cdot k_3 - 3k_4 &= 0 \\ 2k_1 - k_2 + k_3 + 7k_4 &= 0 \\ 4k_1 + 3k_2 + 2k_3 + 2k_4 &= 0 \end{aligned}$$

which is a system of homogeneous linear equation and can be put in the form $AX = 0$.

$$\text{i.e., } \begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Operate $R_2 - 2R_1, R_3 - 4R_1$,

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & -5 & 2 & 14 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Operate $R_3 - R_2$;

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \dots(i)$$

$$\therefore k_1 + 2k_2 - 3k_4 = 0$$

$$-5k_2 + k_3 + 13k_4 = 0$$

$$k_3 + k_4 = 0$$

$$\therefore k_4 = -k_3$$

$$\text{From (ii), } 5k_2 = k_3 - 13k_3 = -12k_3 \therefore k_2 = -\frac{12}{5}k_3$$

$$\text{From (i), } k_1 = +3k_4 - 2k_2 = -3k_3 + \frac{24}{5}k_3 = \frac{9}{5}k_3$$

$$\text{Let } k_3 = t$$

$$\therefore k_1 = \frac{9}{5}t, k_2 = -\frac{12}{5}t, k_3 = t, k_4 = -t$$

$\therefore$ Given vectors are L.D. and the relation between them is $\frac{9}{5}tx_1 - \frac{12}{5}tx_2 + tx_3 - tx_4 = 0$

$$\text{or } 9x_1 - 12x_2 + 5x_3 - 5x_4 = 0.$$

Example 2. Show that the column vectors of the matrix

(P.T.U., Dec. 2002)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 2 \end{bmatrix} \text{ are linearly dependent.}$$

$$\text{Sol. Let } X_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, X_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, X_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Consider the matrix equation

$$k_1 X_1 + k_2 X_2 + k_3 X_3 = 0$$

$$k_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + k_3 \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 0$$

$$\text{i.e., } \begin{aligned} k_1 + 2k_2 + 3k_3 &= 0 \\ -2k_1 + k_2 + 2k_3 &= 0 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = 0$$

$$\text{Operate } R_2 + 2R_1: \begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 8 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = 0$$

$$\text{or } \begin{aligned} k_1 + 2k_2 + 3k_3 &= 0 \\ 5k_2 + 8k_3 &= 0 \end{aligned}$$

$$\therefore k_2 = -\frac{8}{5}k_3; k_1 = -\frac{16}{5}k_3 - 3k_3 = -\frac{1}{5}k_3$$

$$\text{Let } k_3 = \lambda \neq 0$$

$$\therefore k_1 = -\frac{1}{5}\lambda, k_2 = -\frac{8}{5}\lambda, k_3 = \lambda$$

$\therefore$ Given column vectors are L.D.

Sol. Let $X_1 = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$, $X_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $X_3 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

Consider $k_1 X_1 + k_2 X_2 + k_3 X_3 = 0$

i.e., $k_1 \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + k_3 \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} = 0$

$\cdot tx_4 = 0$

or

$$3k_1 + k_2 + k_3 = 0$$

$$2k_1 + 0 \cdot k_2 - k_3 = 0$$

$$4k_1 + 2k_2 - k_3 = 0$$

ec. 2002)

or

$$\begin{bmatrix} 3 & 1 & 1 \\ 2 & 0 & -1 \\ 4 & 2 & -1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Operate $R_1 - R_2$; $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 4 & 2 & -1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Operate $R_2 - 2R_1$, $R_3 - 4R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & -2 & -9 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Operate $R_3 - R_2$;

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \begin{cases} k_1 + k_2 + 2k_3 = 0 \\ -2k_2 - 5k_3 = 0 \\ -4k_3 = 0 \end{cases} \Rightarrow k_3 = 0, k_2 = 0, k_1 = 0$$

$\therefore$ Given vectors are not linearly dependent. These are linearly independent.

1.16. LINEAR TRANSFORMATIONS

Let a point $P(x, y)$ in a plane transform to the point $P'(x', y')$ under reflection in the co-ordinate axes, reflection in the line $y = x \tan \theta$ or rotation of OP through an angle θ about the origin or rotation of

A Textbook of Engineering Mathematics

an angle θ etc. Then the co-ordinates of P' can be expressed in terms of those of P by the linear relations of the form

$$\begin{aligned} x' &= a_1x + b_1y \\ y' &= a_2x + b_2y \end{aligned}$$

which in matrix notation is $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ or $X' = AX$

such transformations are called linear transformation in two dimensions.

Similarly, relations of the form

$$\begin{aligned} x' &= a_1x + b_1y + c_1z \\ y' &= a_2x + b_2y + c_2z \\ z' &= a_3x + b_3y + c_3z \end{aligned}$$

which in matrix notation is $\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ or $X' = AX$ gives a linear transformation

$(x, y, z) \longrightarrow (x', y', z')$ in three dimensions.

In general, the relation $Y = AX$, where $Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$, $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

defines a linear transformation which carries any vector X into another vector Y over the matrix A which is called the linear operator of the transformation.

This transformation is called linear because $Y_1 = AX_1$ and $Y_2 = AX_2$ implies $aY_1 + bY_2 = A(aX_1 + bX_2)$ for all values of a and b .

Thus, if $X = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ then $Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \end{bmatrix}$

so that $(2, -3) \rightarrow (5, -5)$ under the transformation defined by A .

If the transformation matrix A is non-singular, i.e., if $|A| \neq 0$, then the linear transformation is called **non-singular or regular**.

If the transformation matrix A is singular, i.e., if $|A| = 0$, then the linear transformation is also called **singular**.

For a non-singular transformation $Y = AX$, since A is non-singular, A^{-1} exists and we can write the inverse transformation, which carries the vector Y back into the vector X , as $X = A^{-1}Y$.

Note. If a transformation from $(x_1, x_2, \dots, x_n)$ to $(y_1, y_2, \dots, y_n)$ is given by $Y = AX$ and another transformation from $(y_1, y_2, \dots, y_n)$ to $(z_1, z_2, \dots, z_n)$ is given by $Z = BY$, then the transformation from $(x_1, x_2, \dots, x_n)$ to $(z_1, z_2, \dots, z_n)$ is given by $Z = BY = B(AX) = (BA)X$.

1.17(a). ORTHOGONAL TRANSFORMATION

33

The linear transformation $Y = AX$, where

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

is said to be *orthogonal* if it transforms $y_1^2 + y_2^2 + \dots + y_n^2$ into $x_1^2 + x_2^2 + \dots + x_n^2$.

1.17(b). ORTHOGONAL MATRIX

The matrix A of the above transformation is called *an orthogonal matrix*.

Now,
$$X'X = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1^2 + x_2^2 + \dots + x_n^2$$

and similarly
$$Y'Y = y_1^2 + y_2^2 + \dots + y_n^2.$$

$\therefore$ If $Y = AX$ is an orthogonal transformation, then

$$X'X = x_1^2 + x_2^2 + \dots + x_n^2 = y_1^2 + y_2^2 + \dots + y_n^2.$$

$$= Y'Y = (AX)'(AX) = (X'A')(AX)$$

$$= X'(A'A)X$$

$$[\because (AB)' = BA]$$

which holds only when $A'A = I$ or when $A'A = A^{-1}A$

or when $A' = A^{-1}$

$$[\because A^{-1}A = I]$$

Hence a real square matrix A is said to be *orthogonal* if $AA' = A'A = I$

Also, for an orthogonal matrix A , $A' = A^{-1}$.

1.17(c). PROPERTIES OF AN ORTHOGONAL MATRIX

(P.T.U., May 2001, Dec. 2001)

(i) The transpose of an orthogonal matrix is orthogonal.

Proof. Let A be an orthogonal matrix

$$\therefore AA' = I = A'A$$

Taking transpose of both sides of $AA' = I$

$$(AA')' = I' \text{ or } (A')'A' = I \text{ i.e., product of } A' \text{ and its transpose i.e., } (A')' = I$$

$\therefore A'$ is an orthogonal matrix

(P.T.U., Dec. 2001)

(ii) The inverse of an orthogonal matrix is orthogonal

Proof. Let A be an orthogonal matrix $\therefore AA' = I$

Take inverse of both sides $(AA')^{-1} = I^{-1}$

$$\text{or } (A')^{-1}A^{-1} = I \quad \text{or } (A^{-1})'(A^{-1}) = I$$

i.e., Product of A^{-1} and its transpose i.e., $(A^{-1})'$ is I

$\therefore A^{-1}$ is orthogonal.

(iii) If A is an orthogonal matrix then $|A| = \pm 1$

Proof: A is an orthogonal matrix

$$\therefore AA' = I$$

Take determinant of both sides

$$|AA'| = |I| \quad \text{or } |A||A'| = 1 \because |I| = 1$$

$$\text{i.e.,} \quad |A|^2 = 1 \quad \because |A'| = |A|$$

$$\text{i.e.,} \quad |A| = \pm 1$$

Note. An orthogonal matrix A is called proper or improper according as $|A| = 1$ or -1 .

(iv) The product of two orthogonal matrices of the same order is orthogonal

Proof: Let A, B be two orthogonal matrices of the same order so that

$$AA' = BB' = I$$

$$\begin{aligned} \text{Now,} \quad (AB)(AB)' &= (AB)(B'A') = A(BB')A' \\ &= A(I)A' = (AI)A' = AA' = I \end{aligned}$$

$\therefore AB$ is also an orthogonal matrix.

Example 1. Show that the transformation $y_1 = 2x_1 + x_2 + x_3, y_2 = x_1 + x_2 + 2x_3, y_3 = x_1 - 2x_3$ is regular.

Write down the inverse transformation.

(Kerala, 1990 S, 95; Rewa, 1994)

Sol. In matrix notation, the given transformation is $Y = AX$,

$$\text{where} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\text{Since} \quad |A| = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{vmatrix} = -1 \neq 0,$$

the matrix A is non-singular and hence, the given transformation is non-singular or regular.

$$\begin{aligned} \text{Now,} \quad A^{-1} &= \frac{\text{adj } A}{|A|} = \frac{\begin{bmatrix} -2 & 4 & -1 \\ 2 & -5 & 1 \\ 1 & -3 & 1 \end{bmatrix}}{-1} \\ &= -\begin{bmatrix} -2 & 4 & -1 \\ 2 & -5 & 1 \\ 1 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -4 & 1 \\ -2 & 5 & -1 \\ -1 & 3 & -1 \end{bmatrix} \end{aligned}$$

The inverse transformation is given by $X = A^{-1}Y$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & -4 & 1 \\ -2 & 5 & -1 \\ -1 & 3 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\Rightarrow \quad x_1 = 2y_1 - 4y_2 + y_3, \quad x_2 = -2y_1 + 5y_2 - y_3, \quad x_3 = -y_1 + 3y_2 - y_3.$$

Example 2. Prove that the following matrix is orthogonal

$$\frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}.$$

(P.T.U., May 2007; Poona, 1990)

Sol. Denoting the given matrix by A, we have

$$A' = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Now, } AA' &= \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \times \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Since $AA' = I$, A is an orthogonal matrix.

TEST YOUR KNOWLEDGE

- Are the following vectors linearly dependent? If so, find a relation between them.
 - $x_1 = (1, 2, 1), x_2 = (2, 1, 4), x_3 = (4, 5, 6), x_4 = (1, 8, -3)$
 - $x_1 = (2, -1, 4), x_2 = (0, 1, 2), x_3 = (6, -1, 16), x_4 = (4, 0, 12)$
 - $x_1 = (2, -1, 3, 2), x_2 = (1, 3, 4, 2), x_3 = (3, -5, 2, 2)$
 - $x_1 = (2, 3, 1, -1), x_2 = (2, 3, 1, -2), x_3 = (4, 6, 2, 1)$
 - $x_1 = (2, 2, 1)^t, x_2 = (1, 3, 1)^t, x_3 = (1, 2, 2)^t$ where 't' stands for transpose. [Hii]
- Show that the transformation $y_1 = x_1 - x_2 + x_3, y_2 = 3x_1 - x_2 + 2x_3, y_3 = 2x_1 - 2x_2 + 3x_3$ is non-singular inverse transformation.
- Represent each of the transformation $x_1 = 3y_1 + 2y_2, y_1 = z_1 + 2z_2, x_2 = -y_1 + 4y_2$ and $y_2 = z_1 + z_2$ as matrices and find the composite transformation which expresses x_1, x_2 in terms of z_1, z_2 .
(Kerala, 1990, 95)
- A transformation from the variables x_1, x_2, x_3 to y_1, y_2, y_3 is given by $Y = AX$, and another transformation from y_1, y_2, y_3 to z_1, z_2, z_3 is given by $Z = BY$, where $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 1 & 3 & 5 \end{bmatrix}$. Obtain the transformation from x_1, x_2, x_3 to z_1, z_2, z_3 .

- Which of the following matrices are orthogonal

$$(i) \frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 2 & -3 & 1 \\ 4 & 3 & 1 \\ -3 & 1 & 9 \end{bmatrix}$$

- Prove that the following matrix is orthogonal

$$\begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{bmatrix}$$

ANSWERS

1. (i) Yes; $x_3 = 2x_1 + x_2$ and $x_4 = 5x_1 - 2x_2$
 (iii) Yes; $2x_1 - x_2 - x_3 = 0$

(ii) Yes; $x_3 = 3x_1 + 2x_2$ and $x_4 = 2x_1 + x_2$
 (iv) Yes; $5x_1 - 3x_2 - x_3 = 0$. (v) no, L.L.

2. $x_1 = \frac{1}{2}(y_1 + y_2 - y_3)$, $x_2 = \frac{1}{2}(-5y_1 + y_2 + y_3)$, $x_3 = -4y_1 + 2y_3$.

3. $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 & 6 \\ 11 & -2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$

5. (i) Orthogonal (ii) Not orthogonal.

4. $Z = (BA)X$, where $BA = \begin{bmatrix} 1 & 4 & -1 \\ -1 & 9 & -1 \\ -3 & 14 & -1 \end{bmatrix}$

1.18. COMPLEX MATRICES

If all the elements of a matrix are real numbers, then it is called a *real matrix* or a matrix over R . On the other hand, if at least one element of a matrix is a complex number $a + ib$, where a, b are real and $i = \sqrt{-1}$, then the matrix is called a *complex matrix*.

1.19(a). CONJUGATE OF A MATRIX

The matrix obtained by replacing the elements of a complex matrix A by the corresponding conjugate complex numbers is called the *conjugate of the matrix A* and is denoted by $\bar{A}$.

Thus, if

$$A = \begin{bmatrix} 2+3i & -7i \\ 5 & 1-i \end{bmatrix}, \text{ then } \bar{A} = \begin{bmatrix} 2-3i & 7i \\ 5 & 1+i \end{bmatrix}.$$

1.19(b). CONJUGATE TRANSPOSE OF A MATRIX

It is easy to see the *conjugate of the transpose of A* i.e., $(\bar{A})'$ and the *transpose conjugate of A* i.e., $(\bar{A})'$ are equal. Each of them is denoted by A^θ .

Thus

$$(\bar{A})' = (\bar{A})' = A^\theta.$$

1.20. HERMITIAN AND SKEW HERMITIAN MATRIX

(P.T.U., May 2002, 2007)

A square matrix A is said to be **Hermitian** if $A^\theta = A$ i.e., if $A = [a_{ij}]$, then $\bar{a}_{ij} = a_{ji} \forall i, j$ and when $i = j$, then $a_{ii} = \bar{a}_{ii} \Rightarrow a_{ii}$ is purely real i.e., all diagonal elements of a Hermitian matrix are purely real while every other element is the conjugate complex of the element in the transposed position.

For example, $A = \begin{bmatrix} 5 & 2+i & -3i \\ 2-i & -3 & 1-i \\ 3i & 1+i & 0 \end{bmatrix}$ is a Hermitian matrix.

A square matrix A is said to be **Skew Hermitian** if $A^\theta = -A$ i.e., if $A = [a_{ij}]$, then $\bar{a}_{ij} = -a_{ji} \forall i, j$ and when $i = j$, then $\bar{a}_{ii} = -a_{ii}$ i.e., if $a_{ii} = a + ib$ then $\bar{a}_{ii} = a - ib$ and $\bar{a}_{ii} = -a_{ii} \Rightarrow a - ib = -(a + ib) \Rightarrow a = 0$
 $\therefore a_{ii}$ is either purely imaginary or zero.

In a skew Hermitian matrix, the diagonal elements are zero or purely imaginary number of the form $i\beta$, where β is real. Every other element is the negative of the conjugate complex of the element in the transposed position.

For example, $B = \begin{bmatrix} 3i & 1+i & 7 \\ -1+i & 0 & -2-i \\ -7 & 2-i & -i \end{bmatrix}$ is a skew Hermitian matrix.

Note. The following result hold :

- (i) $\overline{\overline{A}} = A$ (ii) $\overline{A+B} = \overline{A} + \overline{B}$ (iii) $\overline{\lambda A} = \overline{\lambda} \overline{A}$
 (iv) $\overline{AB} = \overline{A} \overline{B}$ (v) $(A^\theta)^\theta = A$ (vi) $(A+B)^\theta = A^\theta + B^\theta$
 (vii) $(\lambda A)^\theta = \overline{\lambda} A^\theta$ (viii) $(AB)^\theta = B^\theta A^\theta$.

(P.T.U., May 2002)

1.21. UNITARY MATRIX

A complex square matrix A is said to unitary if $A^\theta A = I$ or we can say $(\overline{A}') A = I$

Taking conjugate of both sides $A' \overline{A} = I$

Incase of real matrices : If A is a real matrix there $\overline{A} = A$; then A will be unitary if $A' \overline{A} = I \Rightarrow A' A = I$ which clearly shows that A is also an orthogonal matrix.

Hence every orthogonal matrix is unitary.

1.22. PROPERTIES OF A UNITARY MATRIX

(a) **Determinant of a Unitary Matrix is of Modulus Unity**

Proof. Let A be a unitary matrix

$$\text{Then } AA^\theta = I$$

Taking determinant of both sides $|AA^\theta| = |I|$

$$\text{or } |A| |\overline{A}'| = 1 \text{ or } |A| |\overline{A}| = 1 \therefore |A'| = |A|$$

$$\text{or } |A|^2 = 1 \text{ hence the result.}$$

(b) **The Product of Two Unitary Matrices of the Same Order is Unitary**

(P.T.U., Dec)

Let A, B be two unitary matrices $\therefore AA^\theta = A^\theta A = I$ and $BB^\theta = B^\theta B = I$

$$\text{Now, } (AB)(AB)^\theta = AB(B^\theta A^\theta) = A(BB^\theta)A^\theta = (AI)A^\theta = AA^\theta = I$$

Hence AB is unitary matrix.

(P.T.U., Dec)

(c) **The Inverse of a Unitary Matrix is Unitary**

Proof. Let A be a unitary matrix $\therefore AA^\theta = A^\theta A = I$

$$AA^\theta = I$$

Take inverse of both sides $(AA^\theta)^{-1} = I^{-1}$ or $(A^\theta)^{-1} A^{-1} = I$

$$\text{or } (A^{-1})^\theta (A^{-1}) = I \therefore A^{-1} \text{ is also unitary.}$$

ILLUSTRATIVE EXAMPLES

Example 1. If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, verify that $A^\theta A$ is a Hermitian matrix.

Sol. $A' = \begin{bmatrix} 2+i & -5 \\ 3 & i \\ -1+3i & 4-2i \end{bmatrix}$

$$A^\theta = \overline{(A')} = \begin{bmatrix} 2-i & -5 \\ 3 & -i \\ -1-3i & 4+2i \end{bmatrix}$$

$$\therefore A^\theta A = \begin{bmatrix} 2-i & -5 \\ 3 & -i \\ -1-3i & 4+2i \end{bmatrix} \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$$

$$= \begin{bmatrix} 30 & 6-8i & -19+17i \\ 6+8i & 10 & -5+5i \\ -19-17i & -5-5i & 30 \end{bmatrix} = B(\text{say})$$

Now, $B' = \begin{bmatrix} 30 & 6+8i & -19-17i \\ 6-8i & 10 & -5-5i \\ -19-17i & -5-5i & 30 \end{bmatrix}$

$$B^\theta = \overline{(B')} = \begin{bmatrix} 30 & 6-8i & -19+17i \\ 6+8i & 10 & -5+5i \\ -19-17i & -5-5i & 30 \end{bmatrix} = B$$

$\Rightarrow B (= A^\theta A)$ is a Hermitian matrix.

Example 2. If A and B are Hermitian, show that $AB - BA$ is skew Hermitian.

Sol. A and B are Hermitian. $\Rightarrow A^\theta = A$ and $B^\theta = B$

Now, $(AB - BA)^\theta = (AB)^\theta - (BA)^\theta$
 $= B^\theta A^\theta - A^\theta B^\theta = BA - AB = -(AB - BA)$

$\Rightarrow AB - BA$ is skew Hermitian.

Example 3. If A is a skew Hermitian matrix, then show that iA is Hermitian.

Sol. A is a skew Hermitian matrix $\Rightarrow A^\theta = -A$

Now, $(iA)^\theta = \bar{i}A^\theta = (-i)(-A) = iA$

$\Rightarrow iA$ is a Hermitian matrix.

Example 4. Show that every square matrix is uniquely expressible as the sum of a Hermitian matrix and a skew Hermitian matrix.

Sol. Let A be any square matrix.

Since $(A + A^\theta)^\theta = A^\theta + (A^\theta)^\theta = A^\theta + A = A + A^\theta$

and $(A - A^\theta)^\theta = A^\theta - (A^\theta)^\theta = A^\theta - A = -(A - A^\theta)$

$\therefore A + A^\theta$ is Hermitian and $A - A^\theta$ is skew Hermitian.

Now, $A = \frac{1}{2}(A + A^\theta) + \frac{1}{2}(A - A^\theta) = P + Q$ (say)

where P is Hermitian and Q is skew Hermitian. Thus, every square matrix can be expressed as the sum of a Hermitian matrix and a skew Hermitian matrix.

To Prove Uniqueness : If possible let A be expressible as sum of another set of a Hermitian and a skew Hermitian matrix (say) $A = R + S$, where R is Hermitian i.e., $R^\theta = R$ and S is skew Hermitian i.e., $S^\theta = -S$.

Now,

Adding, we get

Subtracting, we get

which shows that representation of A is unique.

Example 5. If $N = \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix}$, obtain the matrix $(I-N)(I+N)^{-1}$, and show that it is

(Kerala, 1990; Haripur)

Sol.

$$I-N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$I+N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1+2i \\ -1+2i & 1 \end{bmatrix}$$

$$|I+N| = \begin{vmatrix} 1 & 1+2i \\ -1+2i & 1 \end{vmatrix} = 1 - (4i^2 - 1) = 6 \therefore I+N \text{ is non-singular and } (I+N)^{-1}$$

$$\text{adj}(I+N) = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$(I+N)^{-1} = \frac{1}{|I+N|} \text{adj}(I+N) = \frac{1}{6} \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$\therefore (I-N)(I+N)^{-1} = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix} = A \text{ (say)}$$

$$A' = \frac{1}{6} \begin{bmatrix} -4 & 2-4i \\ -2-4i & -4 \end{bmatrix}$$

$$\overline{(A')} = A^{\theta} = \frac{1}{6} \begin{bmatrix} -4 & 2+4i \\ -2+4i & -4 \end{bmatrix}$$

$$A^{\theta}A = \frac{1}{6} \begin{bmatrix} -4 & 2+4i \\ -2+4i & -4 \end{bmatrix} \frac{1}{6} \begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix} = \frac{1}{36} \begin{bmatrix} 36 & 0 \\ 0 & 36 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$A = (I-N)(I+N)^{-1}$ is unitary.

Example 6. Prove that every Hermitian matrix can be written as $A + iB$ where A is real and symmetric, B is real and skew symmetric.

Sol. Let P be any Hermitian matrix.

Then

$$P^{\theta} = P$$

Consider,

$$P = \frac{P + \bar{P}}{2} + i \frac{P - \bar{P}}{2i} = A + iB, \text{ where}$$

$$A = \frac{P + \bar{P}}{2}, B = \frac{P - \bar{P}}{2i}$$

To prove A and B are real.

We know that

$$z = x + iy \therefore \bar{z} = x - iy, \text{ then } \frac{z + \bar{z}}{2} = 2x \text{ (real)}$$

and

$$\frac{z - \bar{z}}{2i} = \frac{2iy}{2i} = y \text{ (real)}$$

Similarly, we can prove in case of matrices that

$$\frac{P + \bar{P}}{2} \text{ is a real matrix and } \frac{P - \bar{P}}{2i} \text{ is also real}$$

$\therefore A, B$ are real.

To prove A is symmetric

$$A' = \left(\frac{P + \bar{P}}{2} \right)' = \frac{P' + P^{\theta}}{2} = \frac{P' + P}{2} \quad (\because P^{\theta} = P)$$

$$= A.$$

$$\text{Similarly, } B' = \left(\frac{P - \bar{P}}{2i} \right)' = \frac{P' - P^{\theta}}{2i} = \frac{P' - P}{2i} = -\frac{P - P'}{2i} = -B \therefore B \text{ is skew symmetric.}$$

TEST YOUR KNOWLEDGE

1. If $A = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, show that A is a Hermitian matrix and iA is a skew Hermitian matrix.
2. If A is any square matrix, prove that $A + A^{\theta}$, AA^{θ} , $A^{\theta}A$ are all Hermitian and $A - A^{\theta}$ is skew Hermitian.
3. If A, B are Hermitian or skew Hermitian, then so is $A + B$.
4. Show that the matrix $B^{\theta}AB$ is Hermitian or skew Hermitian according as A is Hermitian or skew Hermitian.
5. Prove that $\frac{1}{2} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$ is a unitary matrix. (Andhra, 1990)
6. If A is a Hermitian matrix, then show that iA is a skew Hermitian matrix. (Kerala, 1990)

1.23. CHARACTERISTIC EQUATION, CHARACTERISTIC ROOTS OR EIGEN VALUES, TRACE OF A MATRIX

If A is square matrix of order n , we can form the matrix $A - \lambda I$, where λ is a scalar and I is the unit matrix of order n . The determinant of this matrix equated to zero, i.e.,

$$|A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0 \text{ is called the characteristic equation of A.}$$

On expanding the determinant, the characteristic equation can be written as a polynomial equation of degree n in λ of the form $(-1)^n \lambda^n + k_1 \lambda^{n-1} + k_2 \lambda^{n-2} + \dots + k_n = 0$.

The roots of this equation are called the *characteristic roots or latent roots or eigen-values* of A .

Note. The sum of the eigen-values of a matrix A is equal to trace of A .

[The trace of a square matrix is the sum of its diagonal elements].

1.24. EIGEN VECTORS

Consider the linear transformation $Y = AX$... (1)

which transforms the column vector X into the column vector Y . In practice, we are often required to find those vectors X which transform into scalar multiples of themselves.

Let X be such a vector which transforms into λX (λ being a non-zero scalar) by the transformation (1).

Then $Y = \lambda X$... (2)

From (1) and (2), $AX = \lambda X \Rightarrow AX - \lambda IX = 0 \Rightarrow (A - \lambda I)X = 0$... (3)

This matrix equation gives n homogeneous linear equations

$$\left. \begin{aligned} (a_{11} - \lambda)x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \dots + a_{2n}x_n &= 0 \\ \dots &\dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + (a_{nn} - \lambda)x_n &= 0 \end{aligned} \right\} \dots (4)$$

These equations will have a non-trivial solution only if the coefficient matrix $|A - \lambda I|$ is singular

i.e., if $|A - \lambda I| = 0$

This is the characteristic equation of the matrix A and has n roots which are the eigen-values of A . Corresponding to each root of (5), the homogeneous system (3) has a non-zero solution

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ which is called an } \textit{eigen vector or latent vector}.$$

Note. If X is a solution of (3), then so is kX , where k is an arbitrary constant. Thus, the eigen vector corresponding to an eigen-value is not unique.

1.25. PROPERTIES OF EIGEN VALUES AND EIGEN VECTORS (P.T.U., May 2008)

If λ is an eigen value of A and X be its corresponding eigen vector then we have the following properties:

(i) $\alpha\lambda$ is an eigen value of αA and the corresponding eigen vector remains the same.

$$AX = \lambda X \Rightarrow \alpha(AX) = \alpha(\lambda X) \Rightarrow (\alpha A)X = (\alpha\lambda)X$$

$\therefore \alpha\lambda$ is an eigen value of αA and eigen vector is X .

(ii) λ^m is an eigen value of A^m and corresponding eigen vector remains the same (P.T.U., Dec. 2004)

$$AX = \lambda X \Rightarrow A(AX) = A(\lambda X) \Rightarrow (AA)X = \lambda(AX)$$

$$\therefore A^2X = \lambda(A^2X) = \lambda^2X \Rightarrow \lambda^2 \text{ is an eigen value of } A^2$$

and eigen vector is X .

Pre-multiply successively m times by A we get the result.

(iii) $\lambda - k$ is an eigen-value of $A - kI$ and corresponding eigen vector is X .

$$AX = \lambda X \Rightarrow AX - kIX = \lambda X - kIX$$

or

$$(A - kI)X = (\lambda - k)X \Rightarrow \lambda - k \text{ is the eigen value of } A - kI \text{ and eigen vector is } X.$$

(iv) $\frac{1}{\lambda}$ is an eigen vector of A^{-1} (if it exists) and the corresponding eigen vector is X .

(P.T.U., December 2001, May 2005)

$$AX = \lambda X ; \text{Pre-multiply by } A^{-1}$$

$$A^{-1}(AX) = A^{-1}(\lambda X) \Rightarrow (A^{-1}A)X = \lambda(A^{-1}X)$$

$$\text{or } IX = \lambda(A^{-1}X) \text{ or } A^{-1}X = \frac{1}{\lambda}X$$

$\therefore \frac{1}{\lambda}$ is an eigen value of A^{-1} and eigen vector is X .

(v) $\frac{1}{\lambda - k}$ is an eigen value of $(A - kI)^{-1}$ and corresponding eigen vector is X

$$AX = \lambda X \Rightarrow (A - kI)X = (\lambda - k)X$$

Pre-multiply by $(A - kI)^{-1}$, we get

$$X = (A - kI)^{-1}(\lambda - k)X$$

$$\text{Divide by } \lambda - k, \text{ we get } \frac{1}{\lambda - k}X = (A - kI)^{-1}X$$

$$\therefore (A - kI)^{-1}X = \left(\frac{1}{\lambda - k}\right)X$$

$\therefore \frac{1}{\lambda - k}$ is an eigen value of $(A - kI)^{-1}$ and the eigen vector is X .

(vi) $\frac{|A|}{\lambda}$ is an eigen value of $\text{adj } A$.

(P.T.U., December 2003)

$$AX = \lambda X ; \text{Pre-multiply both sides by } \text{adj } A$$

$$(\text{adj } A)(AX) = (\text{adj } A)\lambda X$$

$$\Rightarrow [(\text{adj } A)A]X = \lambda[(\text{adj } A)X]$$

$$\Rightarrow |A|X = \lambda[(\text{adj } A)X]$$

$$\because A(\text{adj } A) = (\text{adj } A)A = |A|I$$

$$\Rightarrow (\text{adj } A)X = \left[\frac{|A|}{\lambda}\right]X$$

$\Rightarrow \frac{|A|}{\lambda}$ is an eigen value of $\text{adj } A$ and eigen vector is X .

(vii) A and A^T have the same eigen values

$\because$ eigen values of A are given by $|A - \lambda I| = 0$

We know that $|A| = |A^T|$

$$\therefore |A - \lambda I| = |(A - \lambda I)^T| = |A^T - (\lambda I)^T| = |A^T - \lambda I|$$

$\therefore$ eigen values of A and A^T are same.

(viii) For a real matrix A , if $\alpha + i\beta$ is an eigen value, then its conjugate $\alpha - i\beta$ is also an eigen value of A .

Since eigen values of A are given by its characteristic equation $|A - \lambda I| = 0$ and if A is real, then characteristic equation is also a real polynomial equation and in a real polynomial equation, imaginary roots always occur in conjugate pairs. If $\alpha + i\beta$ is an eigen value then $\alpha - i\beta$ is also an eigen value.

ILLUSTRATIVE EXAMPLES

Example 1. (i) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$. (Maa)

(ii) Find the eigen values of the matrix $\begin{bmatrix} 1 & 2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix}$. (P)

9 Sol. (i) The characteristic equation of the given matrix is

$$|A - \lambda I| = 0 \quad \text{or} \quad \begin{vmatrix} 1-\lambda & -2 \\ -5 & 4-\lambda \end{vmatrix} = 0$$

$$\text{or} \quad (1-\lambda)(4-\lambda) - 10 = 0 \quad \text{or} \quad \lambda^2 - 5\lambda - 6 = 0$$

$$\text{or} \quad (\lambda-6)(\lambda+1) = 0 \quad \therefore \lambda = 6, -1.$$

Thus, the eigen values of A are 6, -1

Corresponding to $\lambda = 6$, the eigen vectors are given by $(A - 6I)X = 0$

$$\text{or} \quad \begin{bmatrix} 1-6 & -2 \\ -5 & 4-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad \text{or} \quad \begin{bmatrix} -5 & -2 \\ -5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

we get only one independent equation $-5x_1 - 2x_2 = 0$

$$\therefore \frac{x_1}{2} = \frac{x_2}{-5} \text{ gives the eigen vector } (2, -5)$$

$$\text{Corresponding to } \lambda = -1, \text{ the eigen vectors are given by } \begin{bmatrix} 2 & -2 \\ -5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

We get only one independent equation $2x_1 - 2x_2 = 0$.

$$\therefore x_1 = x_2 \text{ gives the eigen vector } (1, 1).$$

(ii) The characteristic equation of the given matrix is $|A - \lambda I| = 0$

$$\text{i.e., } \begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & -4-\lambda & 2 \\ 0 & 0 & 7-\lambda \end{vmatrix} = 0, \text{ expand w.r.t. 1st column, we get}$$

$$(1-\lambda)(-4-\lambda)(7-\lambda) = 0, \text{ i.e., } \lambda = 1, \lambda = -4, \lambda = 7.$$

Hence the eigen values are -4, 1, 7.

Example 2. Find the eigen values and eigen vectors of the following matrices

$$(i) A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

(P.T.U., Dec. 2001; Delhi, 1997; J.N.T.U., 1995; Coimbatore, 1990;)

Sol. (i) The characteristic equation of the given matrix is $|A - \lambda I| = 0$

$$\text{or } \begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$$

$$\text{or } (-2-\lambda)[- \lambda(1-\lambda)-12]-2[-2\lambda-6]-3[-4+1(1+\lambda)]=0$$

$$\text{or } \lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

By trial, $\lambda = -3$ satisfies it.

$$\therefore (\lambda + 3)(\lambda^2 - 2\lambda - 15) = 0 \Rightarrow (\lambda + 3)(\lambda + 3)(\lambda - 5) = 0 \Rightarrow \lambda = -3, -3, 5$$

Thus, the eigen values of A are -3, -3, 5.

Corresponding to $\lambda = -3$, eigen vectors are given by

$$(A + 3I)X = 0 \quad \text{or} \quad \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

We get only one independent equation $x_1 + 2x_2 - 3x_3 = 0$

Choosing $x_2 = 0$, we have $x_1 - 3x_3 = 0$

$$\therefore \frac{x_1}{3} = \frac{x_2}{0} = \frac{x_3}{1} \text{ giving the eigen vector } (3, 0, 1)$$

Choosing $x_3 = 0$, we have $x_1 + 2x_2 = 0$

$$\therefore \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{0} \text{ giving the eigen vector } (2, -1, 0)$$

Any other eigen vector corresponding to $\lambda = -3$ will be a linear combination of these two.

$$\text{Corresponding to } \lambda = 5, \text{ the eigen vectors are given by } \begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{aligned} -7x_1 + 2x_2 - 3x_3 &= 0 \\ 2x_1 - 4x_2 - 6x_3 &= 0 \\ -x_1 - 2x_2 - 5x_3 &= 0 \end{aligned}$$

$$\begin{array}{ccc|ccc} & x_1 & x_2 & x_3 & & \\ 2 & -3 & -1 & 2 & & \\ -4 & -6 & 2 & -4 & & \end{array}$$

$$\text{From first two equations, we have } \frac{x_1}{-12-12} = \frac{x_2}{-6-42} = \frac{x_3}{28-4}$$

$$\text{or } \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{-1} \text{ giving the eigen vector } (1, 2, -1).$$

(3) (ii) Characteristic equation of the given matrix is

$$|A - \lambda I| = 0, \text{ i.e., } \begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0 \text{ expand w.r.t. } R_1$$

$$(1-\lambda)\{(2-\lambda)(3-\lambda)-2\}-1\{2-2(2-\lambda)\}=0$$

$$\text{or } (1-\lambda)(\lambda^2 - 5\lambda + 4) - 2(\lambda - 1) = 0$$

$$\text{or } (\lambda - 1)[\lambda^2 - 5\lambda + 6] = 0$$

$$\text{or } (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$$

$$\text{or } (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$$

$$\text{i.e., } \lambda = 1, 2, 3.$$

Corresponding to $\lambda = 1$, eigen vectors are given by

$$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

or

$$-x_3 = 0 \Rightarrow x_3 = 0$$

$$x_1 + x_2 + x_3 = 0 \Rightarrow x_1 + x_2 = 0$$

$$2x_1 + 2x_2 + 2x_3 = 0 \text{ i.e., } x_1 = -x_2$$

$$\therefore \frac{x_1}{1} = \frac{x_2}{-1} = \frac{x_3}{0} \therefore \text{eigen vector is } (1, -1, 0)$$

Corresponding to $\lambda = 2$, eigen vectors are given by

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

or

$$-x_1 - x_3 = 0 \Rightarrow x_1 = -x_3$$

$$x_1 + x_3 = 0 \Rightarrow 2x_2 - x_3 = 0$$

$$2x_1 + 2x_2 + x_3 = 0 \text{ or } 2x_2 = x_3$$

$$\therefore \frac{x_1}{-1} = \frac{x_2}{\frac{1}{2}} = \frac{x_3}{1} \text{ or } \frac{x_1}{-2} = \frac{x_2}{1} = \frac{x_3}{2}$$

$\therefore$ Eigen vector is $(-2, 1, 2)$

Corresponding to $\lambda = 3$, eigen vectors are given by

$$\begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

or

$$-2x_1 - x_3 = 0 \therefore x_3 = -2x_1$$

$$x_1 - x_2 + x_3 = 0$$

$$2x_1 + 2x_2 = 0, 2x_1 = -2x_2$$

$$\therefore -x_3 = 2x_1 = -2x_2 \text{ or } \frac{x_1}{1} = \frac{x_2}{-1} = \frac{x_3}{-2}$$

$\therefore$ Eigen vector is $(1, -1, -2)$.

Example 3. Show that if $\lambda_1, \lambda_2, \dots, \lambda_n$ are the latent roots of the matrix A, then A^3 has the latent roots $\lambda_1^3, \lambda_2^3, \dots, \lambda_n^3$.

Sol. Let λ be a latent root of the matrix A. Then there exists a non-zero vector X such that

$$AX = \lambda X \quad \dots(1)$$

$\Rightarrow$

$$A^2(AX) = A^2(\lambda X) \Rightarrow A^3X = \lambda(A^2X)$$

But

$$A^2X = A(AX) = A(\lambda X)$$

$$= \lambda(AX) = \lambda(\lambda X) = \lambda^2X$$

[Using (1)]

$\therefore$

$$A^3X = \lambda(\lambda^2X) = \lambda^3X$$

$\Rightarrow \lambda^3$ is a latent root of A^3 .

$\therefore$ If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the latent roots of A, then $\lambda_1^3, \lambda_2^3, \dots, \lambda_n^3$ are the latent roots of A^3 .

Example 4. Show that eigen values of a Hermitian matrix are real.

Sol. Let A be a Hermitian matrix

$$A^H = A$$

... (1)

Let λ be an eigen value of A then there exists a non-zero vector X such that

$$AX = \lambda X$$

$$(AX)^H = (\lambda X)^H \text{ or } X^H A^H = \bar{\lambda} X^H$$

$$X^H A = \bar{\lambda} X^H$$

[By using (1)]

or

Post multiply both sides by X

$$(X^H A) X = (\bar{\lambda} X^H) X$$

or

$$X^H (AX) = \bar{\lambda} (X^H X)$$

$$X^H (\lambda X) = \bar{\lambda} (X^H X)$$

or

$$\lambda (X^H X) = \bar{\lambda} (X^H X) \Rightarrow \lambda = \bar{\lambda}$$

$\therefore \lambda$ is real.

1.26. CAYLEY HAMILTON THEOREM

(P.T.U., Dec. 2002, May 2004, May 2006, May 2007)

Every square matrix satisfies its characteristic equation.

i.e., if the characteristic equation of the n th order square matrix A is

$$|A - \lambda I| = (-1)^n \lambda^n + k_1 \lambda^{n-1} + k_2 \lambda^{n-2} + \dots + k_n = 0 \quad \dots (1)$$

then

$$(-1)^n A^n + k_1 A^{n-1} + k_2 A^{n-2} + \dots + k_n I = 0 \quad \dots (2)$$

Let

$$P = \text{adj}(A - \lambda I)$$

Since the elements of $A - \lambda I$ are at most of first degree in λ , the elements of $P = \text{adj}(A - \lambda I)$ are polynomials in λ of degree $(n-1)$ or less. We can, therefore, split up P into a number of matrices each containing the same power of λ and write

$$P = P_1 \lambda^{n-1} + P_2 \lambda^{n-2} + \dots + P_{n-2} \lambda^2 + P_{n-1} \lambda + P_n$$

Also, we know that if M is a square matrix, then $M(\text{adj } M) = |M| \times I$

$$\therefore (A - \lambda I)P = |A - \lambda I| \times I$$

By (1) and (2), we have

$$(A - \lambda I)(P_1 \lambda^{n-1} + P_2 \lambda^{n-2} + \dots + P_{n-2} \lambda^2 + P_{n-1} \lambda + P_n) = [(-1)^n \lambda^n + k_1 \lambda^{n-1} + k_2 \lambda^{n-2} + \dots + k_{n-2} \lambda^2 + k_{n-1} \lambda + k_n] I$$

Equating coefficients of like powers of λ on both sides, we have

$$-P_1 = (-1)^n I$$

$$[\because IP_1 = P_1]$$

$$AP_1 - P_2 = k_1 I$$

$$AP_2 - P_3 = k_2 I$$

$$\dots \dots \dots$$

$$AP_{n-2} - P_{n-1} = k_{n-2} I$$

$$AP_{n-1} - P_n = k_{n-1} I$$

$$AP_n = k_n I$$

Pre-multiplying these equations by $A^n, A^{n-1}, A^{n-2}, \dots, A^2, A, I$ respectively and adding, we get

$$0 = (-1)^n A^n + k_1 A^{n-1} + k_2 A^{n-2} + \dots + k_{n-2} A^2 + k_{n-1} A + k_n I \text{ terms on the L.H.S. Cancel in pairs}$$

or

$$(-1)^n A^n + k_1 A^{n-1} + k_2 A^{n-2} + \dots + k_{n-1} A + k_n I = 0$$

... (3)

which proves the theorem.

Matrices

Note 1. Multiplying (3) by A^{-1} , we have $(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I + k_n A^{-1} = O$

$$A^{-1} = -\frac{1}{k_n} [(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I]$$

Thus Cayley Hamilton theorem gives another method for computing the inverse of a matrix. Since this express the inverse of a matrix of order n in terms of $(n-1)$ powers of A , it is most suitable for computing inverse of large matrices.

Note 2. If m be a positive integer such that $m > n$, then multiplying (3) by A^{m-n} , we get

$$(-1)^n A^m + k_1 A^{m-1} + k_2 A^{m-2} + \dots + k_{n-1} A^{m-n+1} + k_n A^{m-n} = O$$

showing that any positive integral power A^m ($m > n$) of A is linearly expressible in terms of those of lower power.

Example 5. Verify Cayley Hamilton theorem for the following matrices :

(i) $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ (P.T.U., May 2006)

(ii) $\begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix}$ (P.T.U., Dec. 2006)

Sol. (i) Let $A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

Characteristic equation of A is $|A - \lambda I| = 0$

or $\begin{vmatrix} 2-\lambda & 3 & 1 \\ 3 & 1-\lambda & 2 \\ 1 & 2 & 3-\lambda \end{vmatrix} = 0$

or $(2-\lambda) \{ (1-\lambda)(3-\lambda) - 4 \} - 3 \{ 3(3-\lambda) - 2 \} + 1 \{ 6 - 1 + \lambda \} = 0$

or $\lambda^3 - 6\lambda^2 - 3\lambda + 18 = 0$

To verify Cayley Hamilton theorem, we have to show that $A^3 - 6A^2 - 3A + 18I = O$

Now, $A^2 = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 14 & 11 & 11 \\ 11 & 14 & 11 \\ 11 & 11 & 14 \end{bmatrix}$

$$A^3 = \begin{bmatrix} 14 & 11 & 11 \\ 11 & 14 & 11 \\ 11 & 11 & 14 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 72 & 75 & 69 \\ 75 & 69 & 72 \\ 69 & 72 & 75 \end{bmatrix}$$

$$A^3 - 6A^2 - 3A + 18I = \begin{bmatrix} 72 & 75 & 69 \\ 75 & 69 & 72 \\ 69 & 72 & 75 \end{bmatrix} - 6 \begin{bmatrix} 14 & 11 & 11 \\ 11 & 14 & 11 \\ 11 & 11 & 14 \end{bmatrix} - 3 \begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 18 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 18 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

(ii) Let $A = \begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix}$

Characteristic equation of A is $|A - \lambda I| = 0$

$$\text{or } \begin{vmatrix} 3-\lambda & 2 & 4 \\ 4 & 3-\lambda & 2 \\ 2 & 4 & 3-\lambda \end{vmatrix} = 0 \text{ or } (3-\lambda) \{(3-\lambda)^2 - 8\} - 2\{12 - 4\lambda - 4\} + 4\{16 - 6 + 2\lambda\} = 0$$

$$\text{or } \lambda^3 - 9\lambda^2 + 3\lambda - 27 = 0$$

To verify Cayley Hamilton theorem, we have to prove

$$A^3 - 9A^2 + 3A - 27I = 0$$

$$A^2 = \begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 28 & 28 \\ 28 & 25 & 28 \\ 28 & 28 & 25 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 25 & 28 & 28 \\ 28 & 25 & 28 \\ 28 & 28 & 25 \end{bmatrix} \begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 243 & 246 & 240 \\ 240 & 243 & 246 \\ 246 & 240 & 243 \end{bmatrix}$$

$$\begin{aligned} A^3 - 9A^2 + 3A - 27I &= \begin{bmatrix} 243 & 246 & 240 \\ 240 & 243 & 246 \\ 246 & 240 & 243 \end{bmatrix} - 9 \begin{bmatrix} 25 & 28 & 28 \\ 28 & 25 & 28 \\ 28 & 28 & 25 \end{bmatrix} + 3 \begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix} - 27 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

Hence theorem is verified.

Example 6. If $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, then use Cayley Hamilton theorem to find the matrix represented by A^5 (P.T.U., May 2001)

Solution. Characteristic equation of A is

$$|A - \lambda I| = 0$$

$$\text{i.e., } \begin{vmatrix} 2-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = 0 \text{ or } \lambda^2 - 7\lambda + 1 = 0$$

By Cayley Hamilton theorem $A^2 - 7A + I = 0$

$$A^2 = 7A - I \quad \dots (1)$$

$$A^4 = 49A^2 - 14A + I$$

$$= 49(7A - I) - 14A + I$$

$$= 329A - 48I$$

$$A^5 = A^4 \cdot A = (329A - 48I)A$$

$$= 329A^2 - 48A = 329(7A - I) - 48A = 2255A - 329I$$

$$= 2255 \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} - 329 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4181 & 6765 \\ 6765 & 10946 \end{bmatrix}$$

Example 7. Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and, hence, find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

Sol. The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By Cayley Hamilton theorem, $A^3 - 5A^2 + 7A - 3I = O$

...(1)

Now, $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

$$= A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + I(A^2 + A + I) = A^2 + A + I \quad [\text{Using (1)}]$$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Example 8. Using Cayley Hamilton Theorem, find the inverse of $\begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ (P.T.U., Dec. 2005)

Solution. Let $A = \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

Characteristic equation of A is $|A - \lambda I| = 0$

$$\text{i.e., } \begin{vmatrix} 2-\lambda & 5 & 3 \\ 3 & 1-\lambda & 2 \\ 1 & 2 & 1-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^3 - 4\lambda^2 - 17\lambda - 4 = 0$$

By Cayley Hamilton theorem,

$$A^3 - 4A^2 - 17A - 4I = 0 \quad \text{or} \quad 4I = A^3 - 4A^2 - 17A$$

Operate both sides by A^{-1} , we get

$$4A^{-1} = A^2 - 4A - 17I$$

$$A^2 = \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 22 & 21 & 19 \\ 11 & 20 & 13 \\ 9 & 9 & 8 \end{bmatrix}$$

$$4A^{-1} = \begin{bmatrix} 22 & 21 & 19 \\ 11 & 20 & 13 \\ 9 & 9 & 8 \end{bmatrix} - 4 \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} - 17 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 1 & 7 \\ -1 & -1 & 5 \\ 5 & 1 & -13 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{4} \begin{bmatrix} -3 & 1 & 7 \\ -1 & -1 & 5 \\ 5 & 1 & -13 \end{bmatrix}$$

TEST YOUR KNOWLEDGE

1. Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

2. Find the eigen values and eigen vectors of the matrices

(i) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

(P.T.U., June 2003, Mysore, 1997 ; Kottayam, 1996 S ; Rewa, 1990 ;
Punjab M.E., 1990 ; Kerala, 1990 S)

(ii) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

(Madras, 1998 ; Kovempu, 1996 ; Rewa 1994 ; Karnataka 1994 ; Poona, 1990 ; Kerala 1990)

(iii) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

(Mysore, 1994 S, 97 S ; Rewa, 1990 ; Madras, 1996)

(iv) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(v) $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

3. Prove that the characteristic roots of a diagonal matrix are the diagonal elements of the matrix.

4. Show that 0 is a characteristic root of a matrix if and only if the matrix is singular.

5. Show that if λ is a characteristic root of the matrix A, then $\lambda + k$ is a characteristic root of the matrix $A + kI$.

6. If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the given values of a matrix A, then A^m has the eigen values $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ (m being a positive integer).
(Kerala, 1990)

7. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$. Show that the equation is satisfied by A and hence obtain the inverse of the given matrix.
(Delhi 1994, 97 ; Madras, 1993 ; Kerala, 1990)

8. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$. Show that the equation is satisfied by A.

9. If $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ use Cayley Hamilton theorem to find A^8 . [Hint: $A^2 = 5I$]

(P.T.U. Dec)

10. Using Cayley-Hamilton theorem, find the inverse of

(i) $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

(ii) $\begin{bmatrix} 7 & -1 & 3 \\ 6 & 1 & 4 \\ 2 & 4 & 8 \end{bmatrix}$

(iii) $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

(P.T.U., Dec)

ANSWERS

1. 1, 6; (1, -1), (1, 4)

2. (i) 0, 3, 15; (1, 2, 2), (2, 1, -2), (2, -2, 1) (ii) 2, 2, 8; (1, 0, -2), (1, 2, 0), (2, -1, 1)

(iii) 1, 2, 3; (1, 0, 1), (1, 0, -1), (0, 1, 0) (iv) 1, 1, 3; (1, -2, 1), (1, 1, 0)

(v) 2, 3, 5; (1, -1, 0), (1, 0, 0), (2, 0, 1)

7. $\lambda^3 - 4\lambda^2 - 20\lambda - 35 = 0$, $\frac{1}{35} \begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ -6 & 1 & -10 \end{bmatrix}$

8. $\lambda^3 - \lambda^2 - 18\lambda - 40 = 0$

9. 6251

10. (i) $\begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} \frac{2}{65} & \frac{2}{13} & -\frac{9}{130} \\ \frac{21}{65} & \frac{5}{13} & \frac{3}{130} \\ \frac{2}{13} & -\frac{3}{13} & \frac{2}{13} \end{bmatrix}$

(iii) $\frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$

1.27. DIAGONALIZABLE MATRICES

A matrix A is said to be diagonalizable if there exists an invertible matrix B . Such that $B^{-1}AB = D$ where D is a diagonal matrix and the diagonal elements of D are the eigen values of A .

Theorem. A square matrix A of order n is diagonalizable if and only if it has n linearly independent vectors.

Proof. Let $X_1, X_2, \dots, X_n$ be n linearly independent eigen vectors corresponding to the eigen values $\lambda_1, \lambda_2, \dots, \lambda_n$ (not necessarily distinct) of matrix A

$\therefore AX_1 = \lambda_1 X_1, AX_2 = \lambda_2 X_2, \dots, AX_n = \lambda_n X_n$

Let $B = [X_1, X_2, \dots, X_n]$ and $D = \text{Diag. } [\lambda_1, \lambda_2, \dots, \lambda_n]$ formed by eigen values

then $AB = A[X_1, X_2, \dots, X_n] = [AX_1, AX_2, \dots, AX_n]$

$= [\lambda_1 X_1, \lambda_2 X_2, \dots, \lambda_n X_n]$

$= [X_1, X_2, \dots, X_n] \text{Diag. } [\lambda_1, \lambda_2, \dots, \lambda_n]$

$AB = BD$

Since columns of B are L.I. $\therefore P(B) = n \therefore B$ is invertible

Pre-multiply both sides by B^{-1}

$\therefore B^{-1}AB = (B^{-1}B)D = D$

$\therefore$ the matrix B , formed by eigen vectors of A , reduces the matrix A to its diagonal form. Post-multiply (1) by B^{-1}

$A(BB^{-1}) = BDB^{-1} \text{ or } A = BDB^{-1}$

Note 1. The matrix B which diagonalises A is called the Modal Matrix of A, obtained by grouping the eigen values of A into a square matrix and matrix D is called Spectral Matrix of A.

Note 2. We have

$$A = BDB^{-1}$$

$$A^2 = A \cdot A = (BDB^{-1})(BDB^{-1}) = BD(B^{-1}B)DB^{-1} \\ = B(DID)B^{-1} = BD^2B^{-1}$$

Repeating this process m times, we get

$$A^m = BD^m B^{-1} \quad (m, a +ve \text{ integer}).$$

$\therefore$ If A is diagonalizable so is A^m .

Note 3. If D is a diagonal matrix of order n and

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}, \text{ then } D^m = \begin{bmatrix} \lambda_1^m & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^m & 0 & \dots & 0 \\ 0 & 0 & \lambda_3^m & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n^m \end{bmatrix}$$

$$A^m = BD^m B^{-1}$$

Similarly if $Q(D)$ is a polynomial in D, then

$$Q(D) = \begin{bmatrix} Q(\lambda_1) & 0 & 0 & \dots & 0 \\ 0 & Q(\lambda_2) & 0 & \dots & 0 \\ 0 & 0 & Q(\lambda_3) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & Q(\lambda_n) \end{bmatrix}$$

$$Q(A) = B[Q(D)]B^{-1}$$

Example 1. Show that the matrix

$$A = \begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

is diagonalizable. Hence find P such that $P^{-1}AP$ is a diagonal matrix, then obtain the matrix $B = A^2 + 5A + 3I$.

Solution. Characteristic equation of A is

[P.T.U. May, 2008]

$$|A - \lambda I| = \begin{vmatrix} 3-\lambda & 1 & -1 \\ -2 & 1-\lambda & 2 \\ 0 & 1 & 2-\lambda \end{vmatrix} = \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

or

$$\lambda = 1, 2, 3.$$

Since the matrix has three distinct eigen values

$\therefore$ it has three linearly independent eigen values and hence A is diagonalizable.

The eigen vector corresponding to $\lambda = 1$ is given by

$$(A - \lambda I)X = \begin{bmatrix} 2 & 1 & -1 \\ -2 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

i.e., $2x_1 + y_1 - z_1 = 0$, $-2x_1 + 2z_1 = 0$ and $y_1 + z_1 = 0$

which gives the solution

$$x_1 = 1, \quad y_1 = -1, \quad z_1 = 1$$

The eigen vector corresponding to $\lambda = 2$ is

$$(A - 2I)X = \begin{bmatrix} 1 & 1 & -1 \\ -2 & -1 & 2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

i.e., $x_1 + y_1 - z_1 = 0$
 $-2x_1 - y_1 + 2z_1 = 0$
 $y_1 = 0$

which gives the solution

$$x_1 = 1, \quad y_1 = 0, \quad z_1 = 1$$

Eigen vector corresponding to $\lambda = 3$ is given by

$$(A - 3I)X = \begin{bmatrix} 0 & 1 & -1 \\ -2 & -2 & 2 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

i.e., $y_1 - z_1 = 0$
 $-2x_1 - 2y_1 + 2z_1 = 0$
 $y_1 - z_1 = 0$

which gives the solution

$$x_1 = 0, \quad y_1 = 1, \quad z_1 = 0$$

$\therefore$ This Modal Matrix

$$P = [X_1, X_2, X_3]$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Now,

$$P^{-1} = \frac{\text{Adj } P}{|P|} = \frac{1}{1} \begin{bmatrix} -1 & -1 & 1 \\ 2 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \text{diag } [1, 2, 3]$$

Hence A is diagonalizable and its diagonal form matrix contains the eigen values only as its diagonal elements.

Now to obtain $B = A^2 + 5A + 3I$ we use $Q(A) = P[Q(D)]P^{-1}$

$$D = \text{diag}(1, 2, 3)$$

$$D^2 = \text{diag}(1, 4, 9)$$

$$A^2 + 5A + 3I = P(D^2 + 5D + 3I)P^{-1}$$

$$\begin{aligned} D^2 + 5D + 3I &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 9 \\ 0 & 0 & 9 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 15 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 0 & 0 \\ 0 & 17 & 0 \\ 0 & 0 & 27 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \therefore B = A^2 + 5A + 3I &= \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 17 & 0 \\ 0 & 0 & 27 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1 \\ 2 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 25 & 8 & -8 \\ -18 & 9 & 18 \\ -2 & 8 & 19 \end{bmatrix} \end{aligned}$$

Example 2. Find a matrix P which transforms the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ into a diagonal form.

(P.T.U., Dec. 2003)

Solution. Let $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

Characteristic equation of A is $|A - \lambda I| = 0$

$$\text{i.e., } \begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0 \text{ or } \lambda^3 - 7\lambda^2 + 36 = 0$$

or $(\lambda + 2)(\lambda^2 - 9\lambda + 18) = 0$

or $(\lambda + 2)(\lambda - 3)(\lambda - 6) = 0$ i.e., $\lambda = -2, 3, 6$ are the eigen values.

When $\lambda = -2$; eigen vectors are given by

$$3x_1 + y_1 + 3z_1 = 0$$

$$x_1 + 7y_1 + z_1 = 0$$

$$3x_1 + y_1 + 3z_1 = 0$$

Solving first and second equations (3^{rd} is same as first)

$$\frac{x_1}{-20} = \frac{y_1}{0} = \frac{z_1}{20} \quad \therefore X_1 = k(-1, 0, 1)$$

Matrices

When $\lambda = 3$; eigen vectors are given by

$$2x_1 + y_1 + 3z_1 = 0$$

$$x_1 + 2y_1 + z_1 = 0$$

$$3x_1 + y_1 - 2z_1 = 0$$

Solving first and second equations :

$$\frac{x_1}{-5} = \frac{y_1}{5} = \frac{z_1}{-5} \therefore X_2 = k(-1, 1, -1)$$

When $\lambda = 6$; eigen vectors are given by

$$-5x_1 + y_1 + 3z_1 = 0$$

$$x_1 - y_1 + z_1 = 0$$

$$3x_1 + y_1 - 5z_1 = 0$$

Solving first and second equations

$$\frac{x_1}{4} = \frac{y_1}{8} = \frac{z_1}{4} \therefore X_3 = k(1, 2, 1)$$

Modal Matrix

$$P = [X_1 \quad X_2 \quad X_3] = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$P^{-1} = \frac{\text{Adj. } P}{|P|} = \frac{-1}{6} \begin{bmatrix} 3 & 2 & -1 \\ 0 & -2 & -2 \\ -3 & 2 & -1 \end{bmatrix} = -\frac{1}{6} \begin{bmatrix} 3 & 0 & -3 \\ 2 & -2 & 2 \\ -1 & -2 & -1 \end{bmatrix}$$

$\therefore$ Required diagonal form $D = P^{-1}AP$

$$= -\frac{1}{6} \begin{bmatrix} 3 & 0 & -3 \\ 2 & -2 & 2 \\ -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$= -\frac{1}{6} \begin{bmatrix} 12 & 0 & 0 \\ 0 & -18 & 0 \\ 0 & 0 & -36 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} \text{ which is formed by the eigen values of } A.$$

10
Example 3. Diagonalise the matrix $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ and obtain its Modal Matrix.

(P.T.U., December)

Solution. Let $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

Characteristic equation of A is $|A - \lambda I| = 0$ i.e., $\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$

or

or

or

or

 $\therefore$

$$\begin{aligned} (-2-\lambda)[- \lambda(1-\lambda)-12] - 2[-2\lambda-6] - 3[-4+1-\lambda] &= 0 \\ -(2+\lambda)(\lambda^2-\lambda-12) + 4(\lambda+3) + 3(\lambda+3) &= 0 \\ -(2+\lambda)(\lambda+3)(\lambda-4) + 7(\lambda+3) &= 0 \\ (\lambda+3)(-\lambda^2+2\lambda+8+7) = 0 \text{ or } -(\lambda+3)(\lambda+3)(\lambda-5) &= 0 \\ \lambda = -3, -3, 5. \end{aligned}$$

Characteristic vectors corresponding to $\lambda = -3$ are

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Operating $R_2 - 2R_1$, $R_3 + R_1$, we get

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \therefore x_1 + 2x_2 - 3x_3 = 0$$

 $\therefore$

$$x_1 = -2x_2 + 3x_3$$

$$x_2 = 1x_2 + 0x_3$$

$$x_3 = 0x_2 + 1x_3$$

 $\therefore$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

 $\therefore$

Eigen vectors are $X_1 = (-2, 1, 0)$ and $X_2 = (3, 0, 1)$.

Characteristic vector corresponding to $\lambda = 5$ is

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \text{ or } \begin{bmatrix} -1 & -2 & -5 \\ -7 & 2 & -3 \\ 2 & -4 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \text{ by operating } R_{22} \text{ and } R_{31}$$

Operate $R_2 - 7R_1$, $R_3 + 2R_1$, we get

$$\begin{bmatrix} -1 & -2 & -5 \\ 0 & 16 & 32 \\ 0 & -8 & -16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Operating $R_3 + \frac{1}{2} R_2$, we get

$$\begin{bmatrix} -1 & -2 & -5 \\ 0 & 16 & 32 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\therefore -x_1 - 2x_2 - 5x_3 = 0$$

$$16x_2 + 32x_3 = 0$$

$$\text{or } x_1 + 2x_2 + 5x_3 = 0$$

$$x_2 + 2x_3 = 0$$

$$\text{or } x_1 = -x_3, \quad x_2 = -2x_3$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = -x_3 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\therefore \text{Eigen vector } X_3 = (1, 2, -1)$$

$$\therefore \text{Modal Matrix } P = \begin{bmatrix} -2 & 3 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}; |P| = 8$$

$$P^{-1} = \frac{\text{Adj. } P}{|P|} = \frac{1}{8} \begin{bmatrix} -2 & 4 & 6 \\ 1 & 2 & 5 \\ 1 & 2 & -3 \end{bmatrix}$$

Now Diagonal Matrix $D = P^{-1}AP$

$$= \frac{1}{8} \begin{bmatrix} -2 & 4 & 6 \\ 1 & 2 & 5 \\ 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} -2 & 2 & 3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} -2 & 3 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$$

$$= \frac{1}{8} \begin{bmatrix} -24 & 0 & 0 \\ 0 & -24 & 0 \\ 0 & 0 & 40 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \text{ i.e., the diagonal matrix formed by eigen values of } A$$